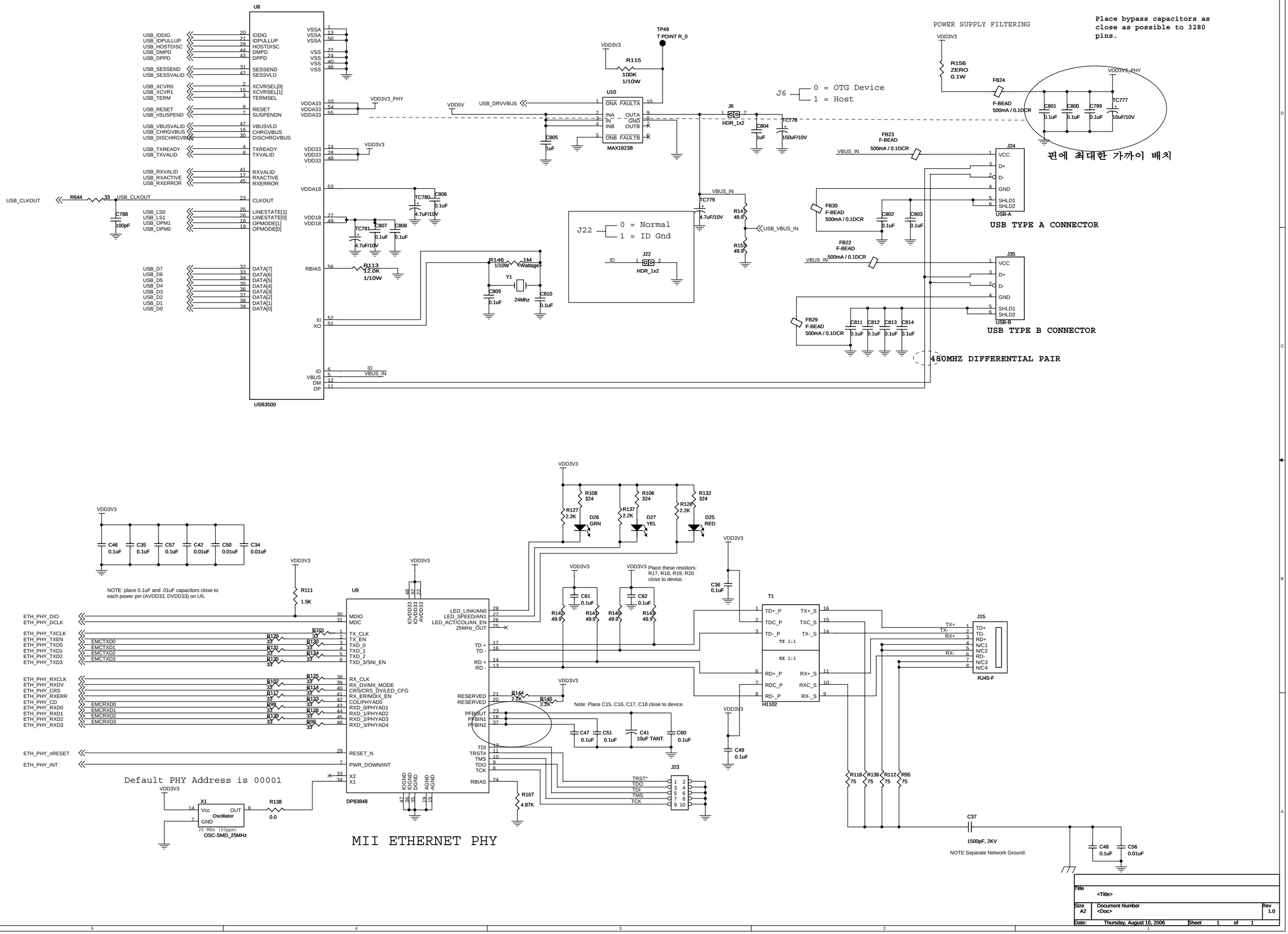
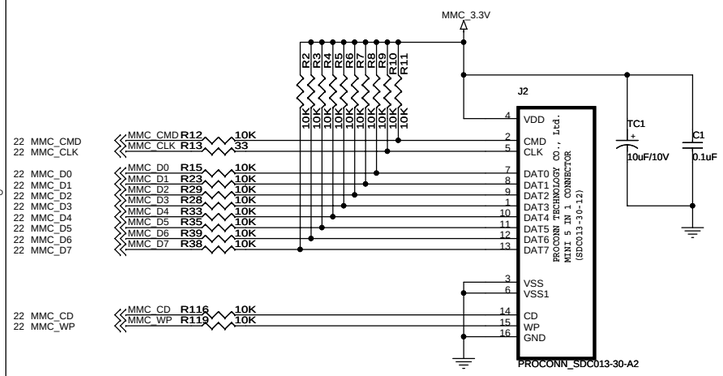
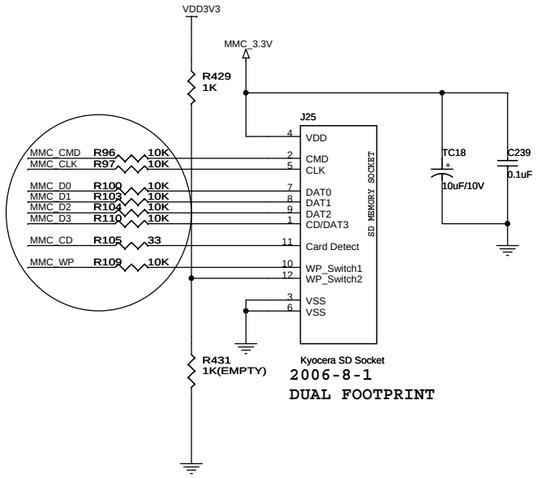
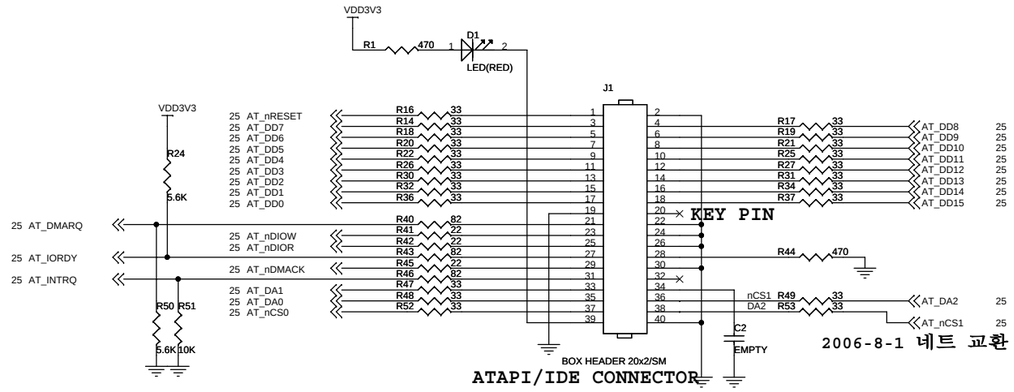
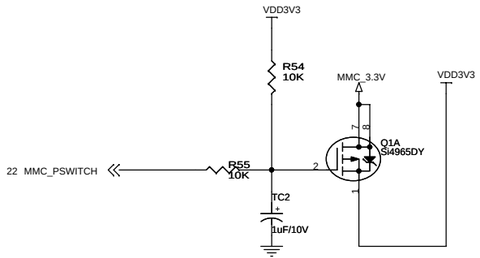

U8
USB3500
U10
MAX1823B
TP49
T POINT R_0
R115
100K
1/10W
J6
HDR_1x2
J6
0 = OTG Device
1 = Host
POWER SUPPLY FILTERING
R156
ZERO
0.1W
FB24
F-BEAD
500mA / 0.1DCR
VDD3V3_PHY
C801
C800
C799
TC777
10uF/10V
Place bypass capacitors as close as possible to 3280 pins.
핀에 최대한 가까이 배치
FB23
F-BEAD
500mA / 0.1DCR
J24
USB-A
USB TYPE A CONNECTOR
FB30
F-BEAD
500mA / 0.1DCR
FB22
F-BEAD
500mA / 0.1DCR
J35
USB-B
USB TYPE B CONNECTOR
FB29
F-BEAD
500mA / 0.1DCR
480MHZ DIFFERENTIAL PAIR
J22
0 = Normal
1 = ID Gnd
HDR_1x2
R146
1M
Y1
24Mhz
R113
12.0K
1/10W
R644
33
USB_CLKOUT
C788
100pF
VDD3V3
VDD5V
USB_DRVVBUS
VBUS_IN
USB_VBUS_IN
TC779
4.7uF/10V
R147
49.9
R153
49.9
TC778
150uF/10V
C804
1uF
C805
1uF
TC780
4.7uF/10V
TC781
4.7uF/10V
C806
C807
C808
0.1uF
C809
C810
USB_IDDIG
USB_IDPULLUP
USB_HOSTDISC
USB_DMPD
USB_DPPD
USB_SESSEND
USB_SESSVALID
USB_XCVR0
USB_XCVR1
USB_TERM
USB_RESET
USB_nSUSPEND
USB_VBUSVALID
USB_CHRGVBUS
USB_DISCHRGVBUS
USB_TXREADY
USB_TXVALID
USB_RXVALID
USB_RXACTIVE
USB_RXERROR
USB_LS0
USB_LS1
USB_OPM1
USB_OPM0
USB_D7
USB_D6
USB_D5
USB_D4
USB_D3
USB_D2
USB_D1
USB_D0
VDD3V3_PHY
ID
VBUS_IN
DM
DP
XI
XO
RBIAS
VDDA18
VDD18
VDD33
VDDA33
VSSA
VSS
NOTE: place 0.1uF and .01uF capacitors close to each power pin (AVDD33, DVDD33) on U5.
C46
C35
C57
C42
C50
C34
0.1uF
0.01uF
R111
1.5K
U9
DP83848
MDIO
MDC
TX_CLK
TX_EN
TXD_0
TXD_1
TXD_2
TXD_3/SNI_EN
RX_CLK
RX_DV/MII_MODE
CRS/CRS_DV/LED_CFG
RX_ER/MDIX_EN
COL/PHYAD0
RXD_0/PHYAD1
RXD_1/PHYAD2
RXD_2/PHYAD3
RXD_3/PHYAD4
RESET_N
PWR_DOWN/INT
X2
X1
LED_LINK/AN0
LED_SPEED/AN1
LED_ACT/COL/AN_EN
25MHz_OUT
TD +
TD -
RD +
RD -
RESERVED
PFBOUT
PFBIN1
PFBIN2
TDI
TRST#
TMS
TDO
TCK
RBIAS
ETH_PHY_DIO
ETH_PHY_DCLK
ETH_PHY_TXCLK
ETH_PHY_TXEN
ETH_PHY_TXD0
ETH_PHY_TXD1
ETH_PHY_TXD2
ETH_PHY_TXD3
ETH_PHY_RXCLK
ETH_PHY_RXDV
ETH_PHY_CRS
ETH_PHY_RXERR
ETH_PHY_CD
ETH_PHY_RXD0
ETH_PHY_RXD1
ETH_PHY_RXD2
ETH_PHY_RXD3
ETH_PHY_nRESET
ETH_PHY_INT
EMCTXD0
EMCTXD1
EMCTXD2
EMCTXD3
EMCRXD0
EMCRXD1
EMCRXD2
EMCRXD3
Default PHY Address is 00001
X1
Oscillator
OSC-SMD_25MHz
R138
0.0
R107
4.87K
R108
324
R106
324
R132
324
R127
2.2K
R137
2.2K
R128
2.2K
D26
GRN
D27
YEL
D25
RED
Place these resistors: R17, R18, R19, R20 close to device.
C61
C62
0.1uF
R142
R141
R140
R143
49.9
R144
2.2K
R145
2.2K
Note: Place C15, C16, C17, C18 close to device.
C47
C51
C41
10uF TANT.
C60
C36
C49
J23
TRST*
TDO
TDI
TMS
TCK
T1
H1102
TX 1:1
RX 1:1
TD+_P
TDC_P
TD-_P
RD+_P
RDC_P
RD-_P
TX+_S
TXC_S
TX-_S
RX+_S
RXC_S
RX-_S
J15
RJ45-F
TD+
TD-
RD+
N/C1
N/C2
RD-
N/C3
N/C4
TX+
TX-
RX+
RX-
R118
R136
R112
R95
75
C37
1500pF, 2KV
NOTE:Separate Network Ground.
C48
C56
0.01uF
MII ETHERNET PHY
Title
<Title>
Size
A2
Document Number
<Doc>
Rev
1.0
Date:
Thursday, August 10, 2006
Sheet
1
of
1

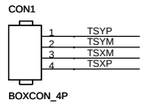




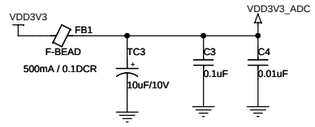


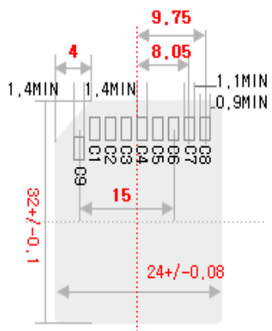


SD/HS-MMC SOCKET

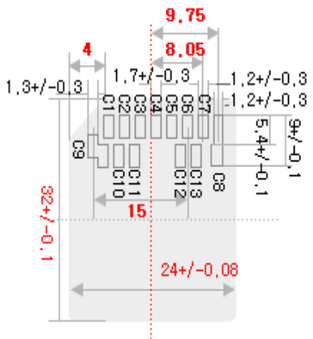


TOUCHSCREEN CONN.
MITSHMI
M60-04-30-134P

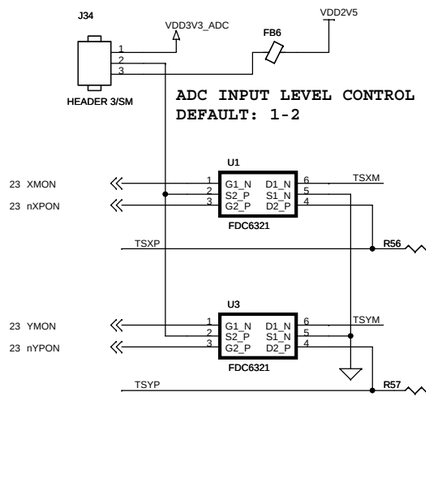


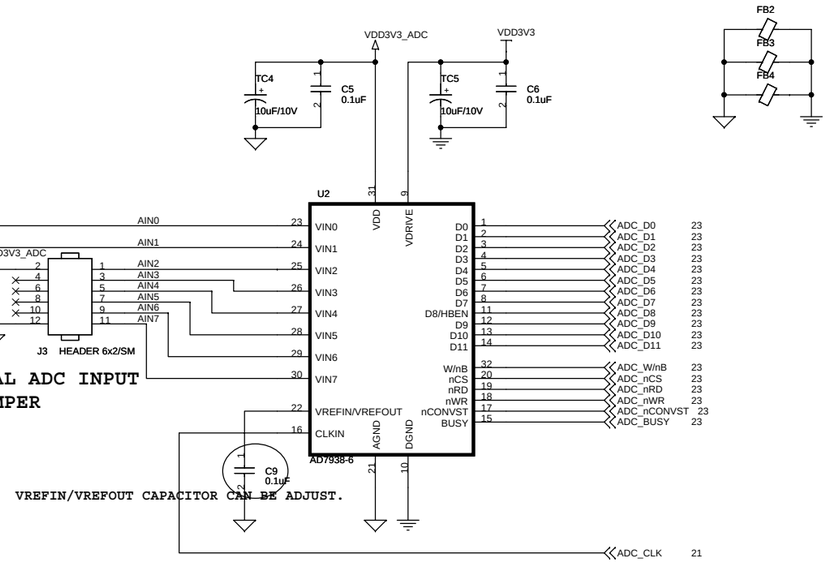


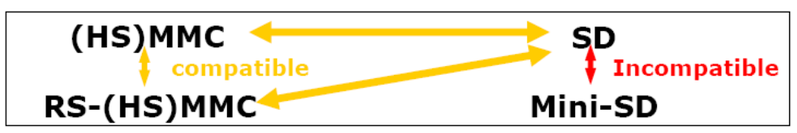
SD Card

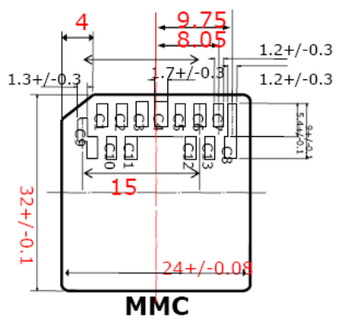


HS-MMC

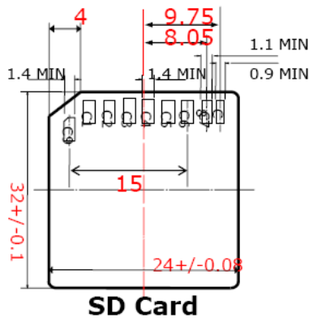








MMC



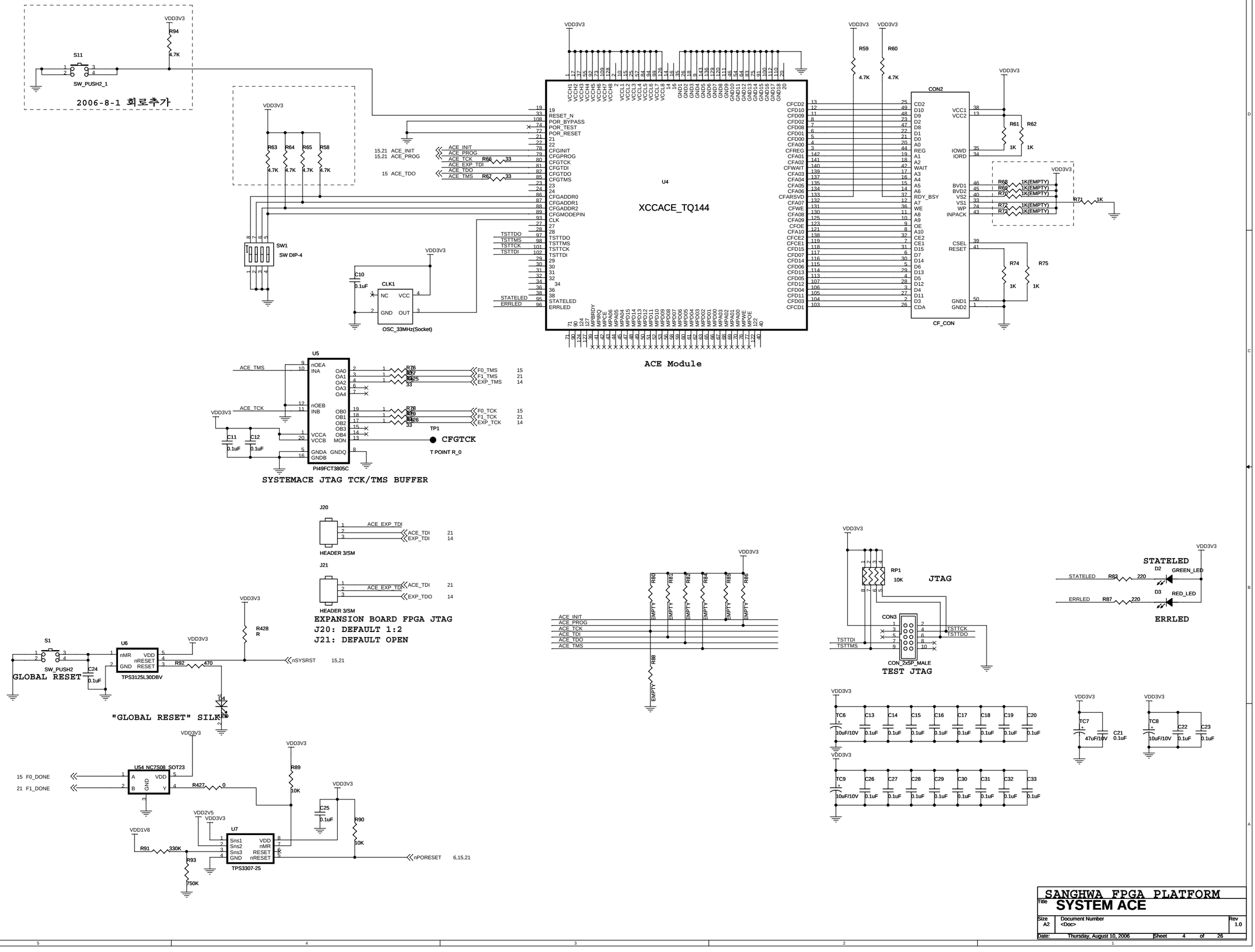
SD Card

SANGHWA FPGA PLATFORM

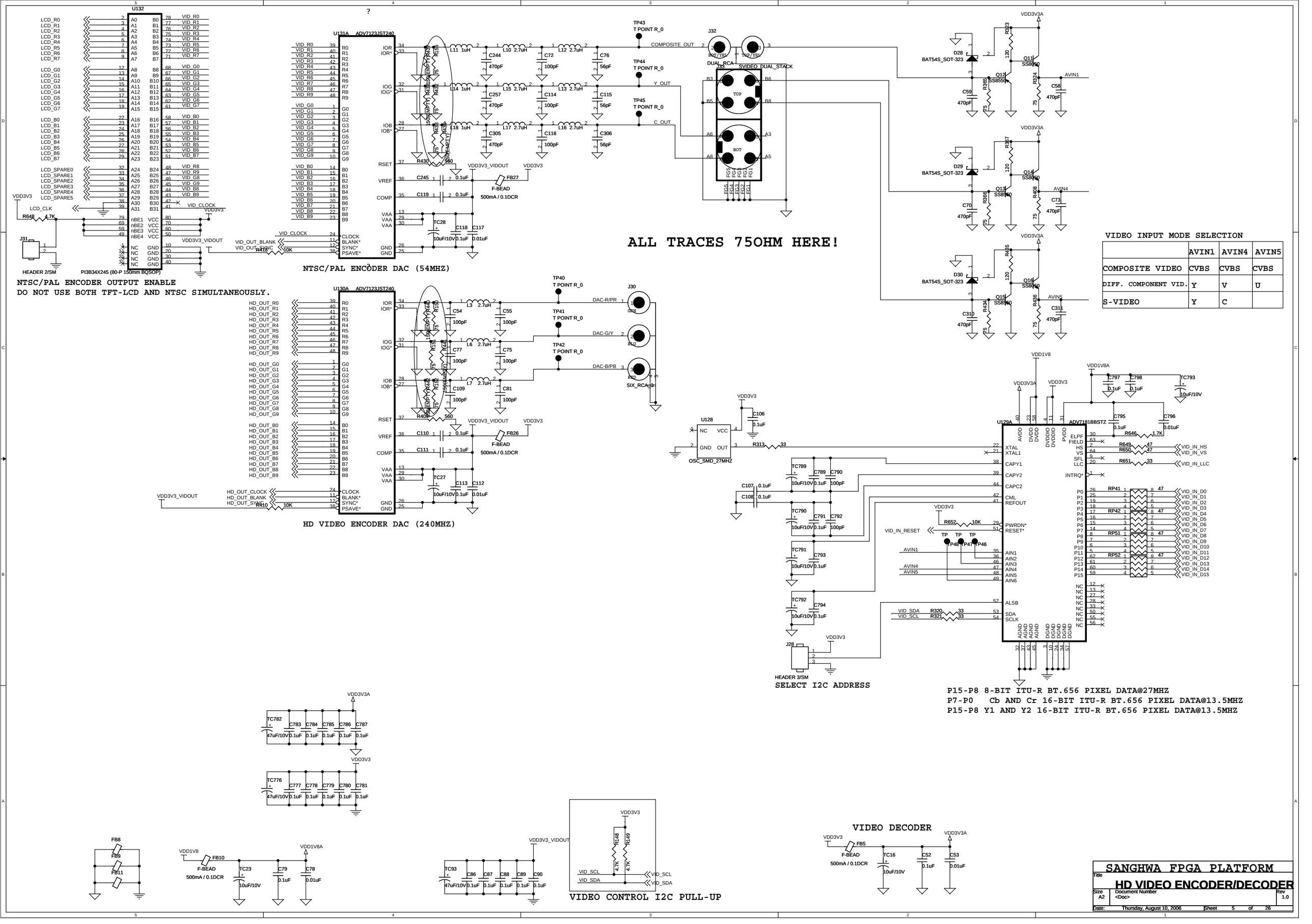
Title: SD/MMC/IDE/TOUCH SCREEN

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Thursday, August 10, 2006 Sheet 3 of 26

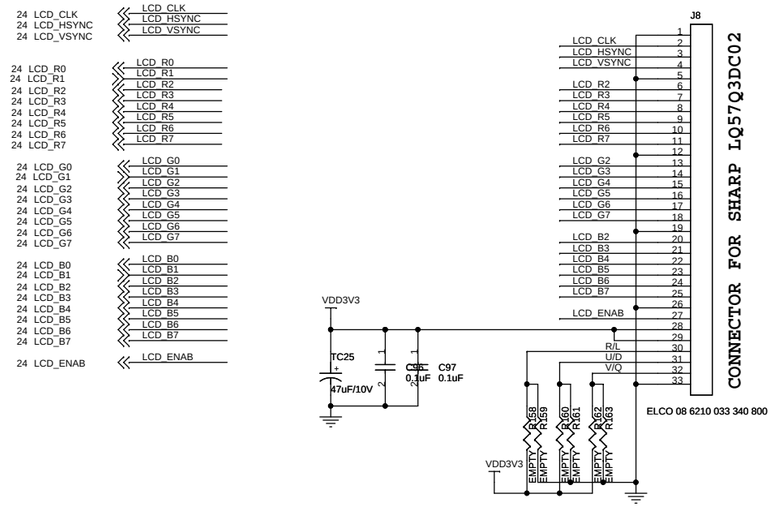


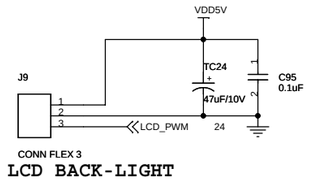
Title	SANGHWA FPGA PLATFORM SYSTEM ACE		
Size A2	Document Number <Doc>		Rev 1.0
Date: Thursday, August 10, 2006	Sheet 4 of 26		



VIDEO INPUT MODE SELECTION

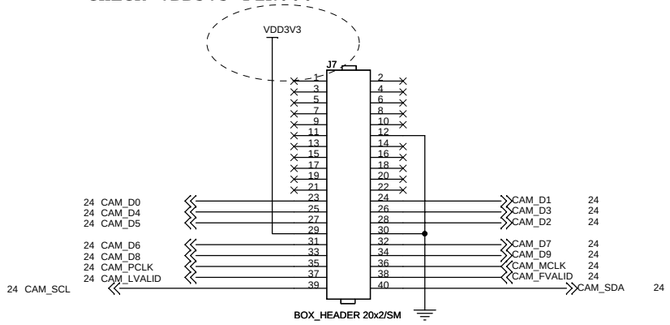
	AVIN1	AVIN4	AVIN5
COMPOSITE VIDEO	CVBS	CVBS	CVBS
DIFF. COMPONENT VID.	Y	V	U
S-VIDEO	Y	C	





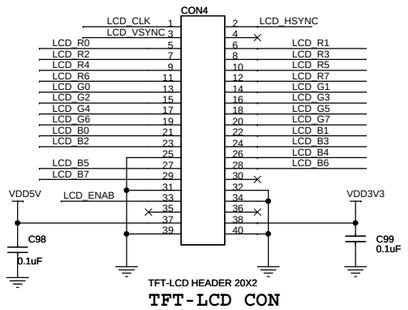
LCD BACK-LIGHT

CHECK VDD3V3 PIN!!!

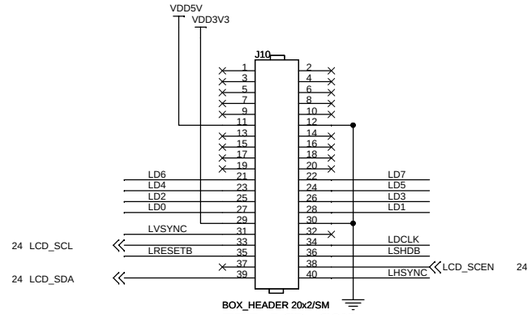


CMOS IMAGE SENSOR SOCKET

TERASIC, INC. (TRDB_DC2 BOARD)

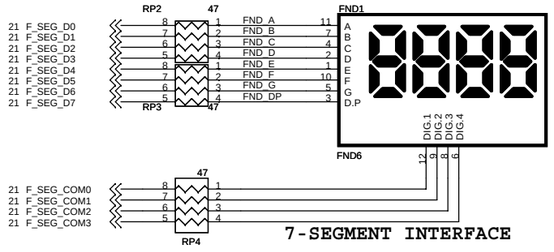


TFT-LCD CON

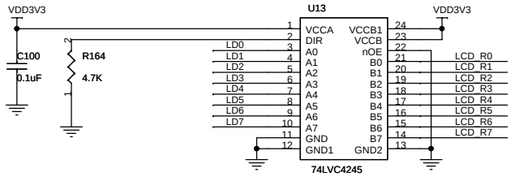


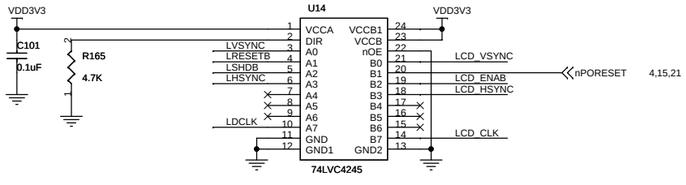
TFT-LCD CON

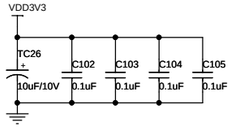
TERASIC, INC. (TRDB_LCM BOARD)



7-SEGMENT INTERFACE





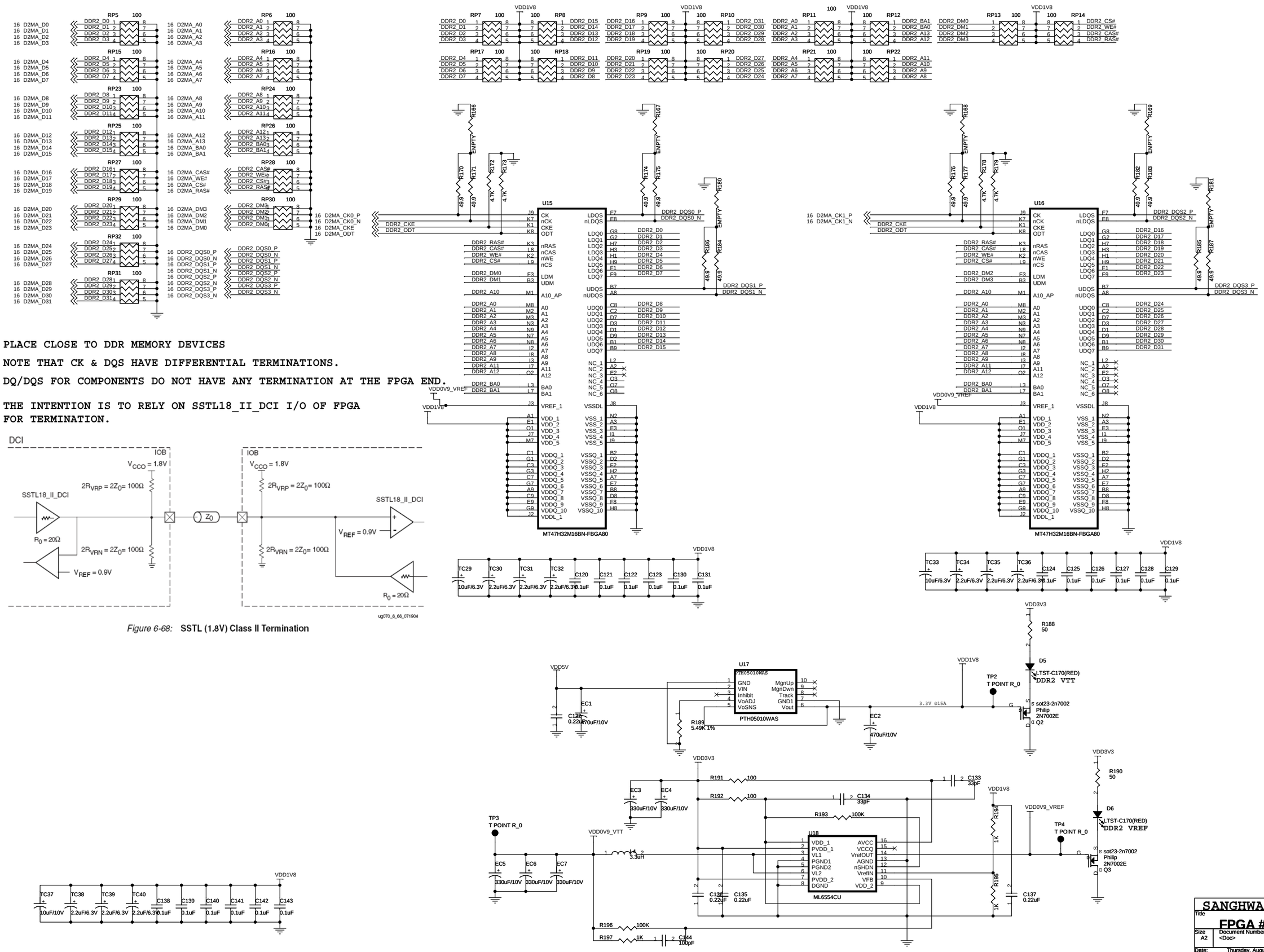


SANGHWA FPGA PLATFORM

Title: CIS/TFT-LCD

Size: A2 | Document Number: <Doc> | Rev: 1.0

Date: Thursday, August 10, 2006 | Sheet 6 of 26



PLACE CLOSE TO DDR MEMORY DEVICES

NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

THE INTENTION IS TO RELY ON SSTL18_II_DCI I/O OF FPGA FOR TERMINATION.

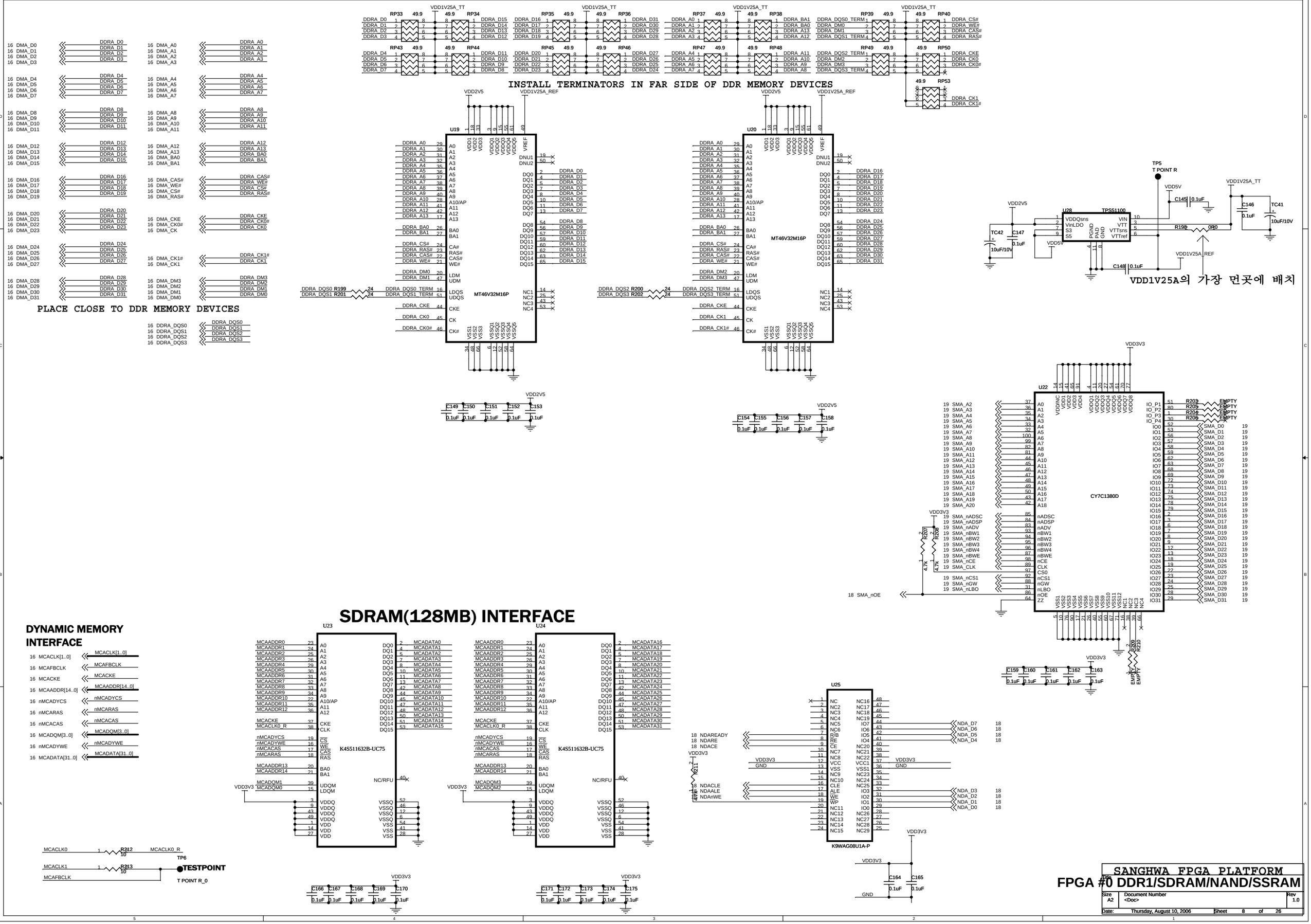
Figure 6-68: **SSTL (1.8V) Class II Termination**

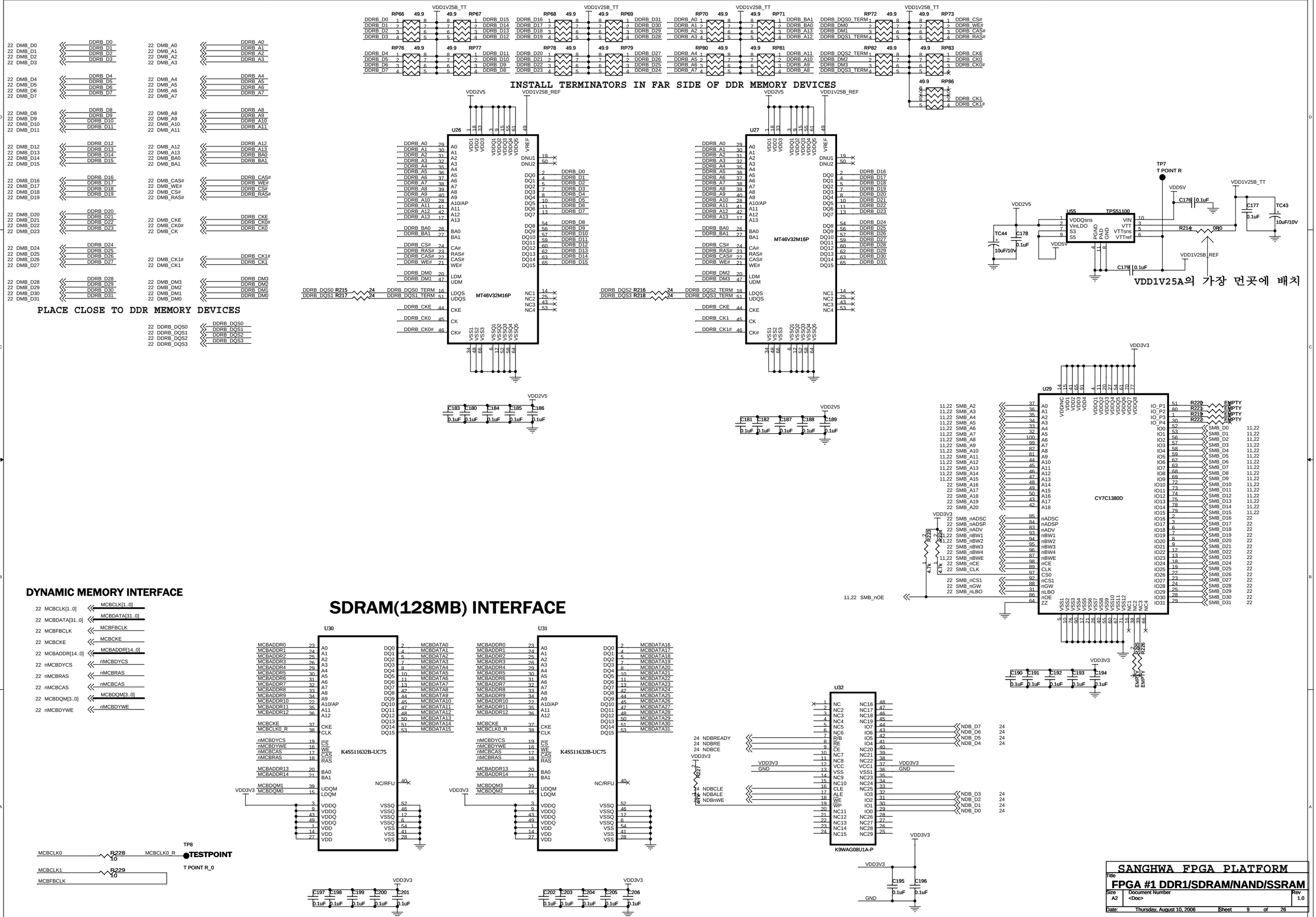
SANGHWA FPGA PLATFORM

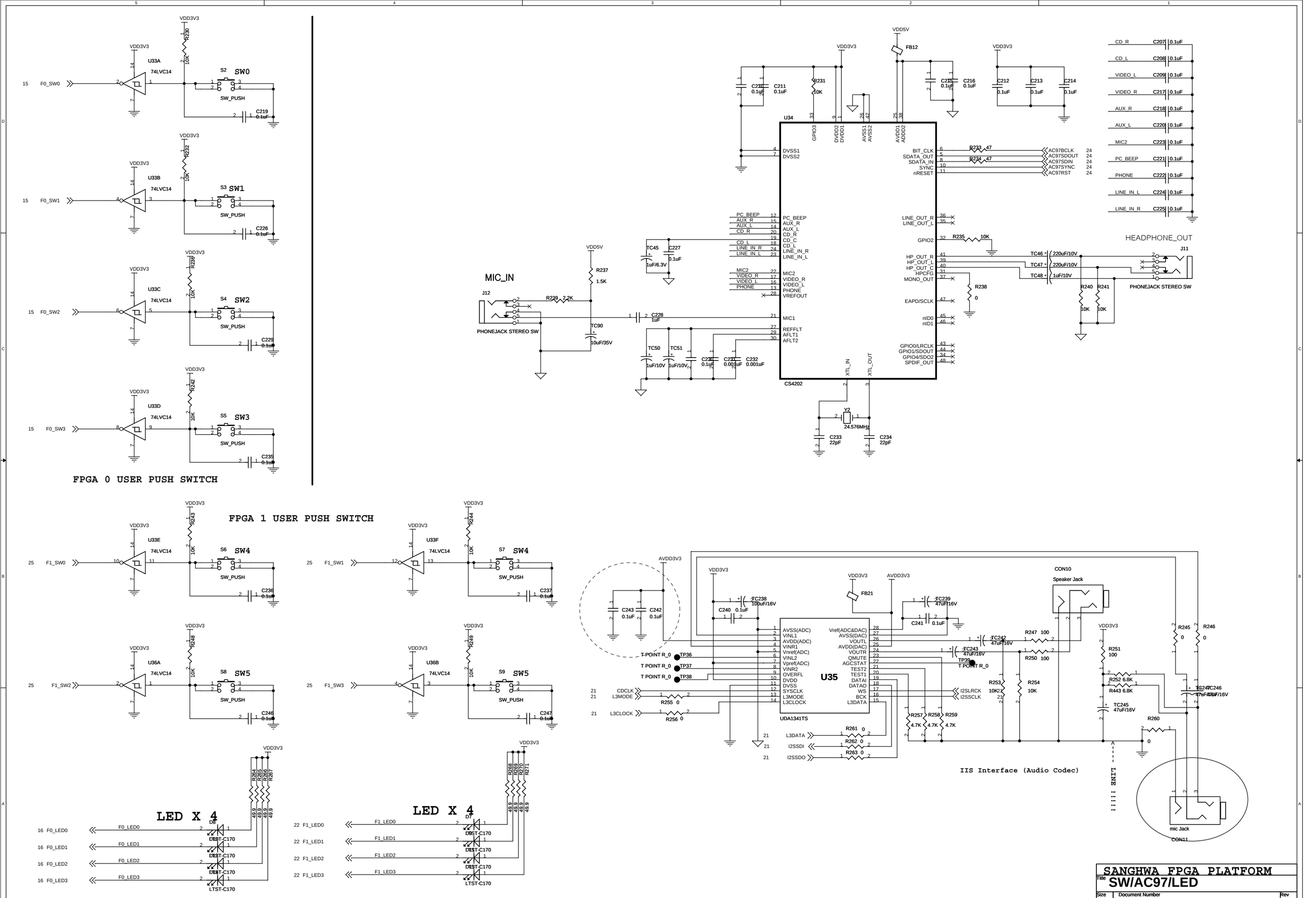
Title: **FPGA #0 DDR2 SDRAM**

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Thursday, August 10, 2006 Sheet 7 of 26


INSTALL TERMINATORS IN FAR SIDE OF DDR MEMORY DEVICES
PLACE CLOSE TO DDR MEMORY DEVICES
VDD1V25A의 가장 먼곳에 배치
SDRAM(128MB) INTERFACE
DYNAMIC MEMORY INTERFACE
U19 MT46V32M16P
U20 MT46V32M16P
U22 CY7C1380D
U23 K4S511632B-UC75
U24 K4S511632B-UC75
U25 K9WAG08U1A-P
U28 TPS51100
TP5 T POINT R
TP6 TESTPOINT T POINT R_0
SANGHWA FPGA PLATFORM
FPGA #0 DDR1/SDRAM/NAND/SSRAM
Size A2
Document Number <Doc>
Rev 1.0
Date: Thursday, August 10, 2006
Sheet 8 of 26


INSTALL TERMINATORS IN FAR SIDE OF DDR MEMORY DEVICES
PLACE CLOSE TO DDR MEMORY DEVICES
VDD1V25A의 가장 먼곳에 배치
DYNAMIC MEMORY INTERFACE
SDRAM(128MB) INTERFACE
U26
U27
MT46V32M16P
U55
TPS51100
U29
CY7C1380D
U30
U31
K4S511632B-UC75
U32
K9WAG08U1A-P
TP7
T POINT R
TP8
TESTPOINT
T POINT R_0
SANGHWA FPGA PLATFORM
FPGA #1 DDR1/SDRAM/NAND/SSRAM
Document Number
<Doc>
Rev
1.0
Size
A2
Date: Thursday, August 10, 2006
Sheet 9 of 26

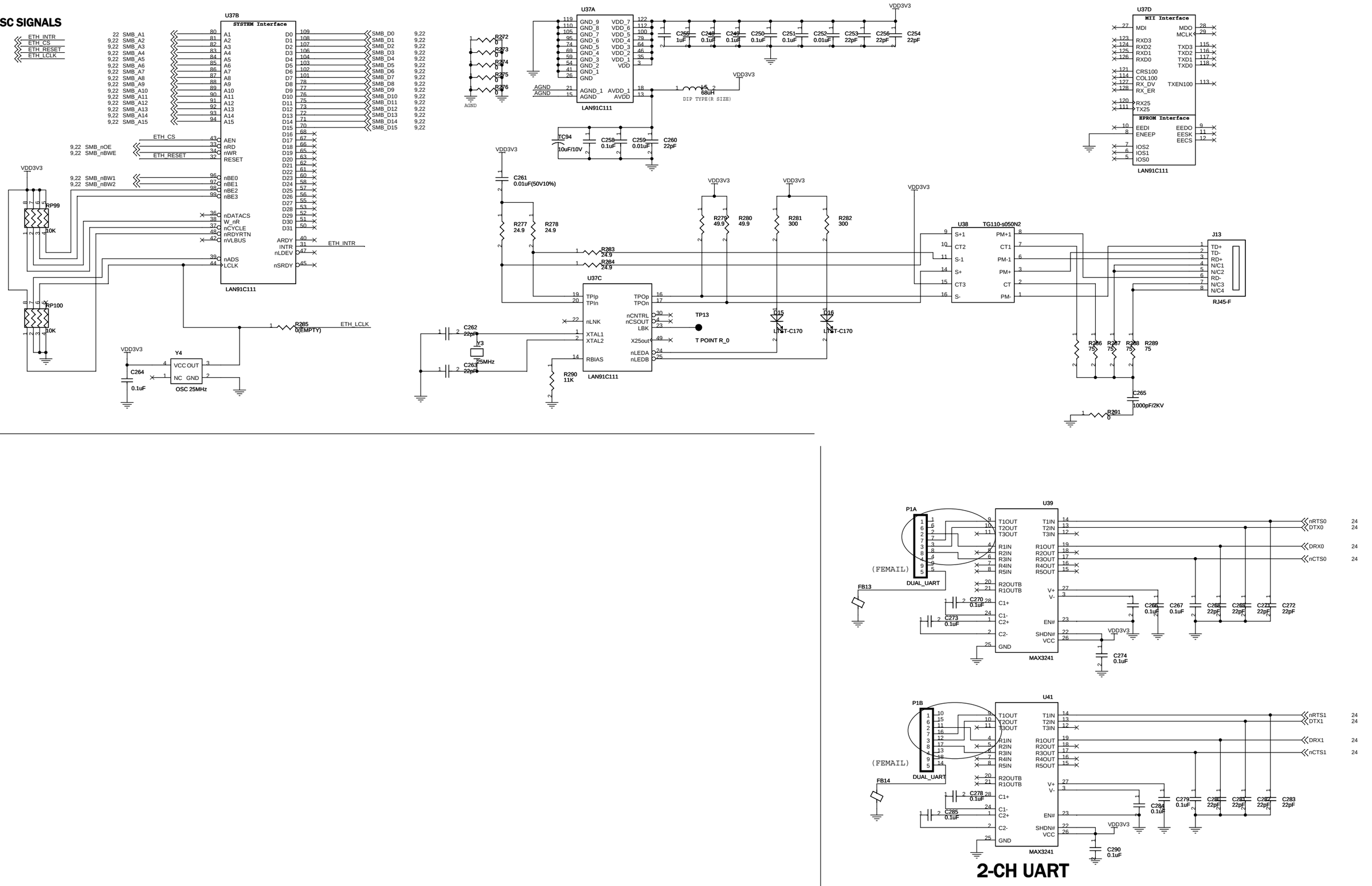


SANGHWA FPGA PLATFORM

Title: SW/AC97/LED

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Thursday, August 10, 2006 Sheet 10 of 26

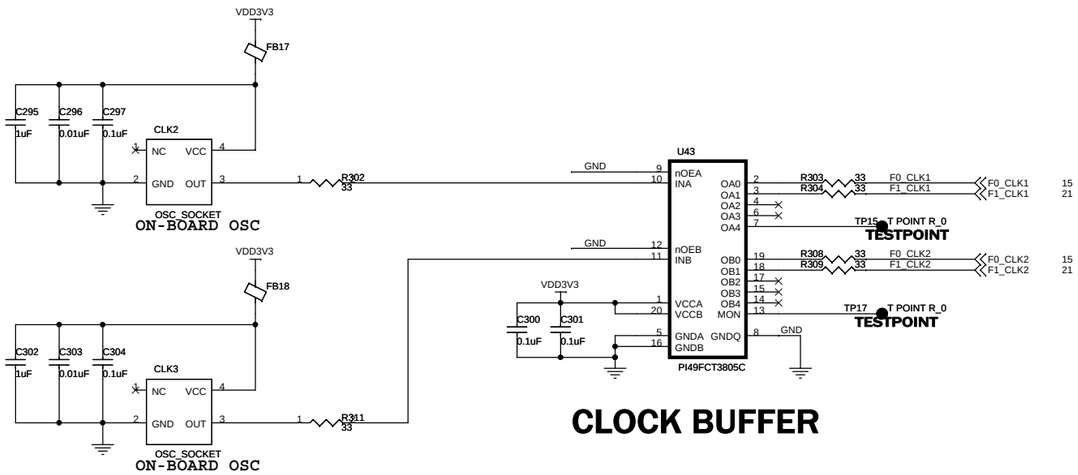


SANGHWA FPGA PLATFORM

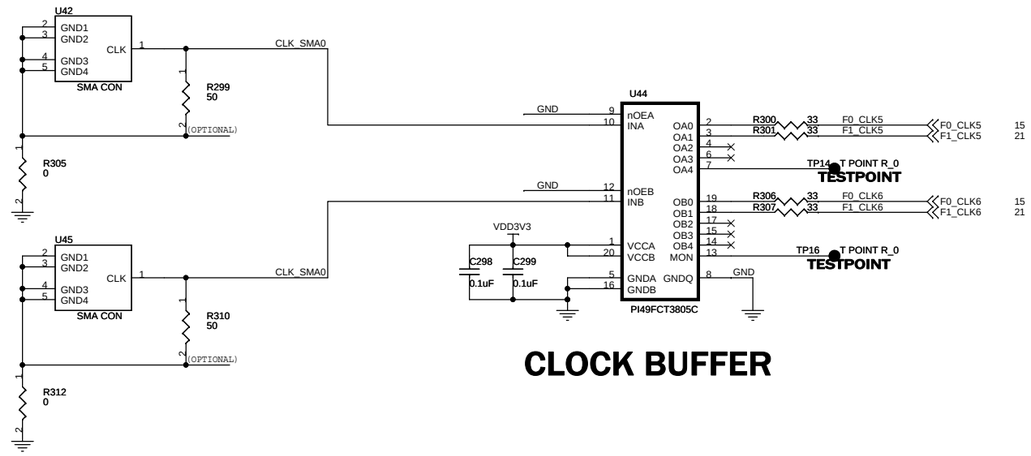
Title **ETH/USB/UART**

Size	Document Number	Rev
A2	<Doc>	1.0

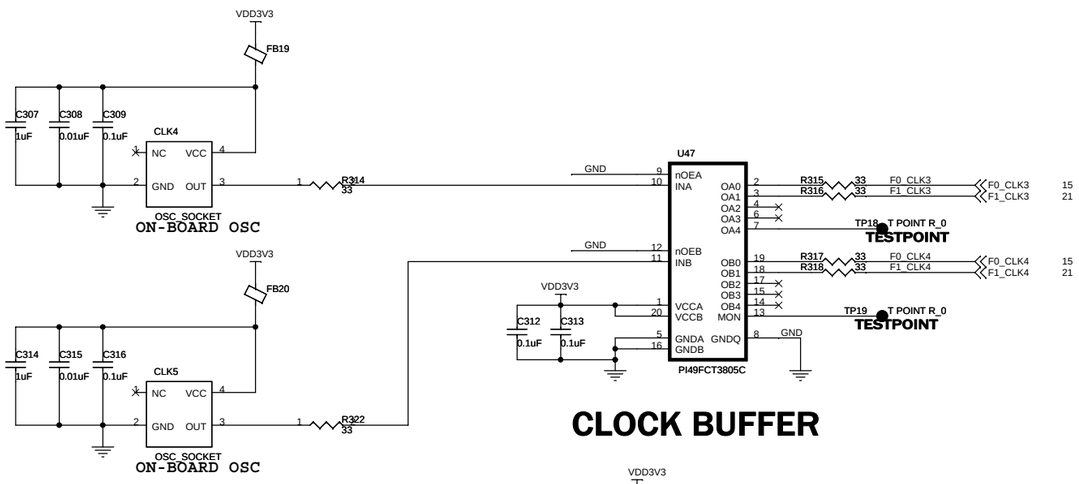
Date: Thursday, August 10, 2006 Sheet 11 of 26



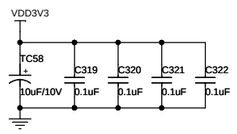
CLOCK BUFFER



CLOCK BUFFER



CLOCK BUFFER

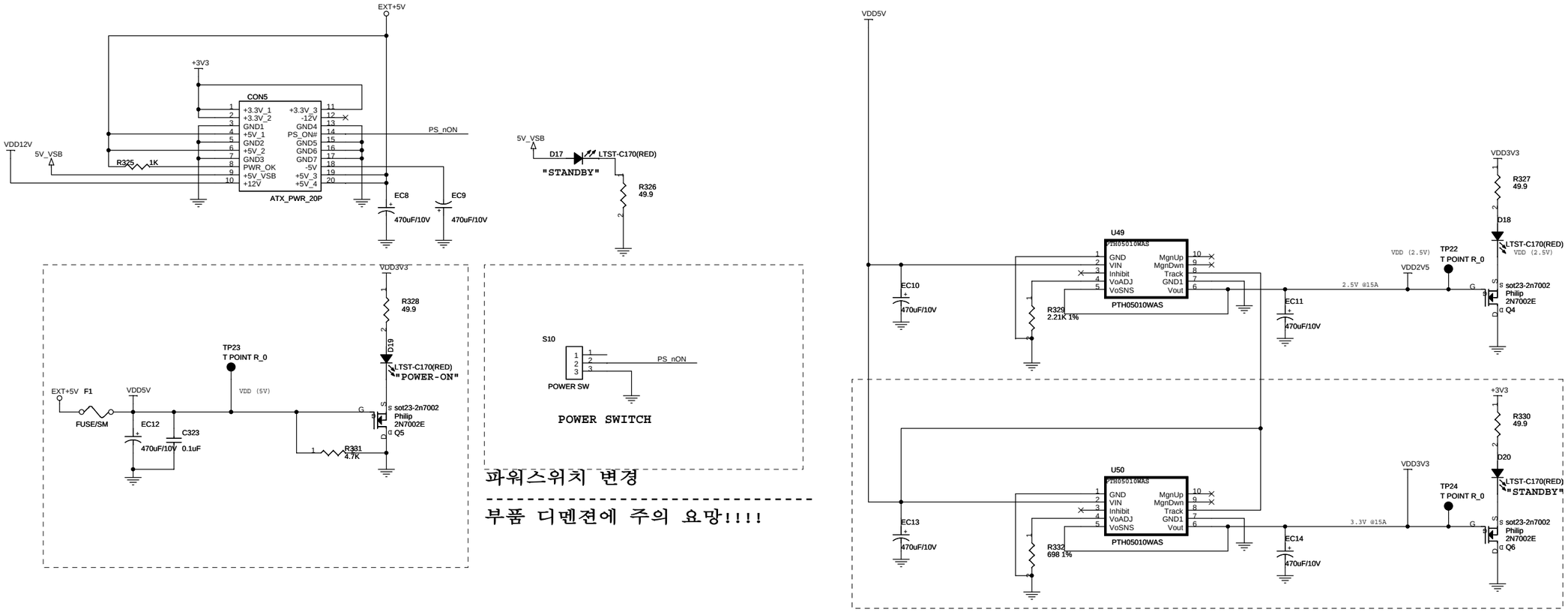


SANGHWA FPGA PLATFORM

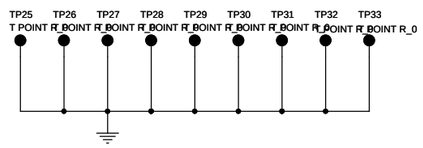
Title **CLOCK/PLL/MULTIPLEXOR**

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Thursday, August 10, 2006 Sheet 12 of 26



GND Point



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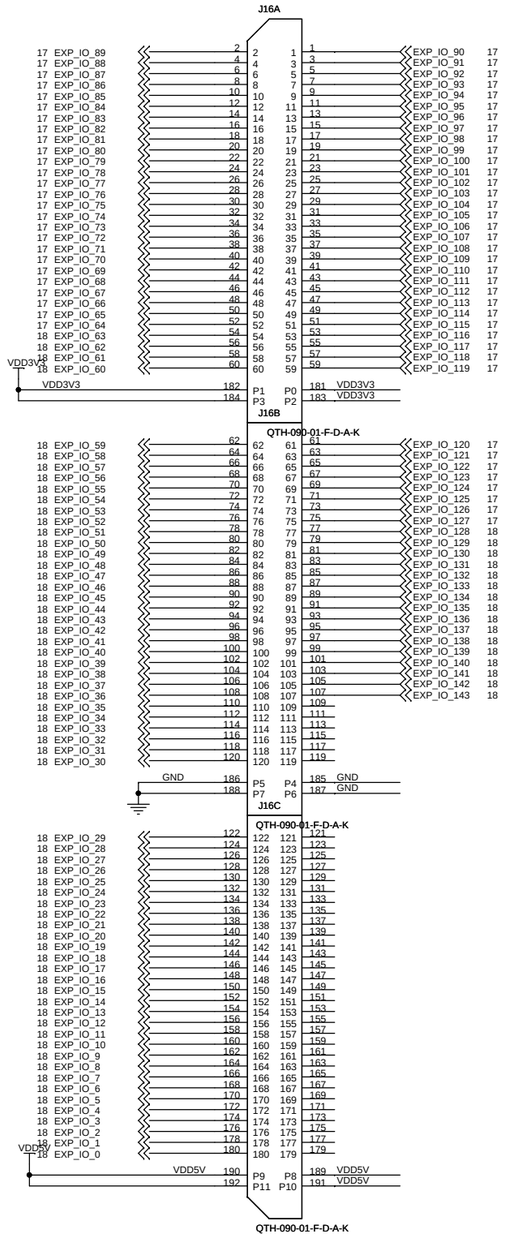
Size A2	Document Number <Doc>	Rev 1.0
Date: Thursday, August 10, 2006	Sheet 13 of 26	

EXPANSION SLOT FOR FPGA #0



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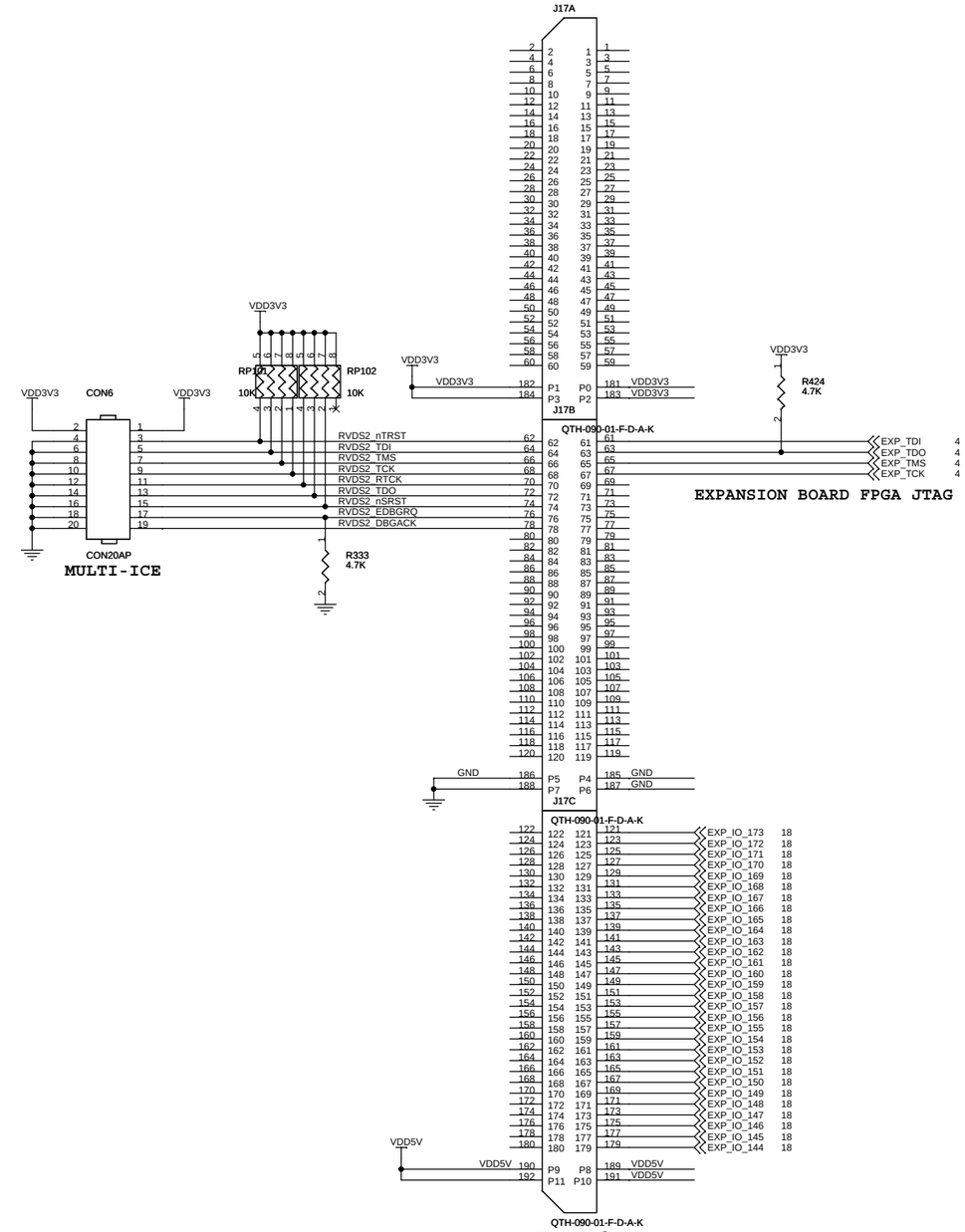

EXP1

SANGHWA FPGA PLATFORM		
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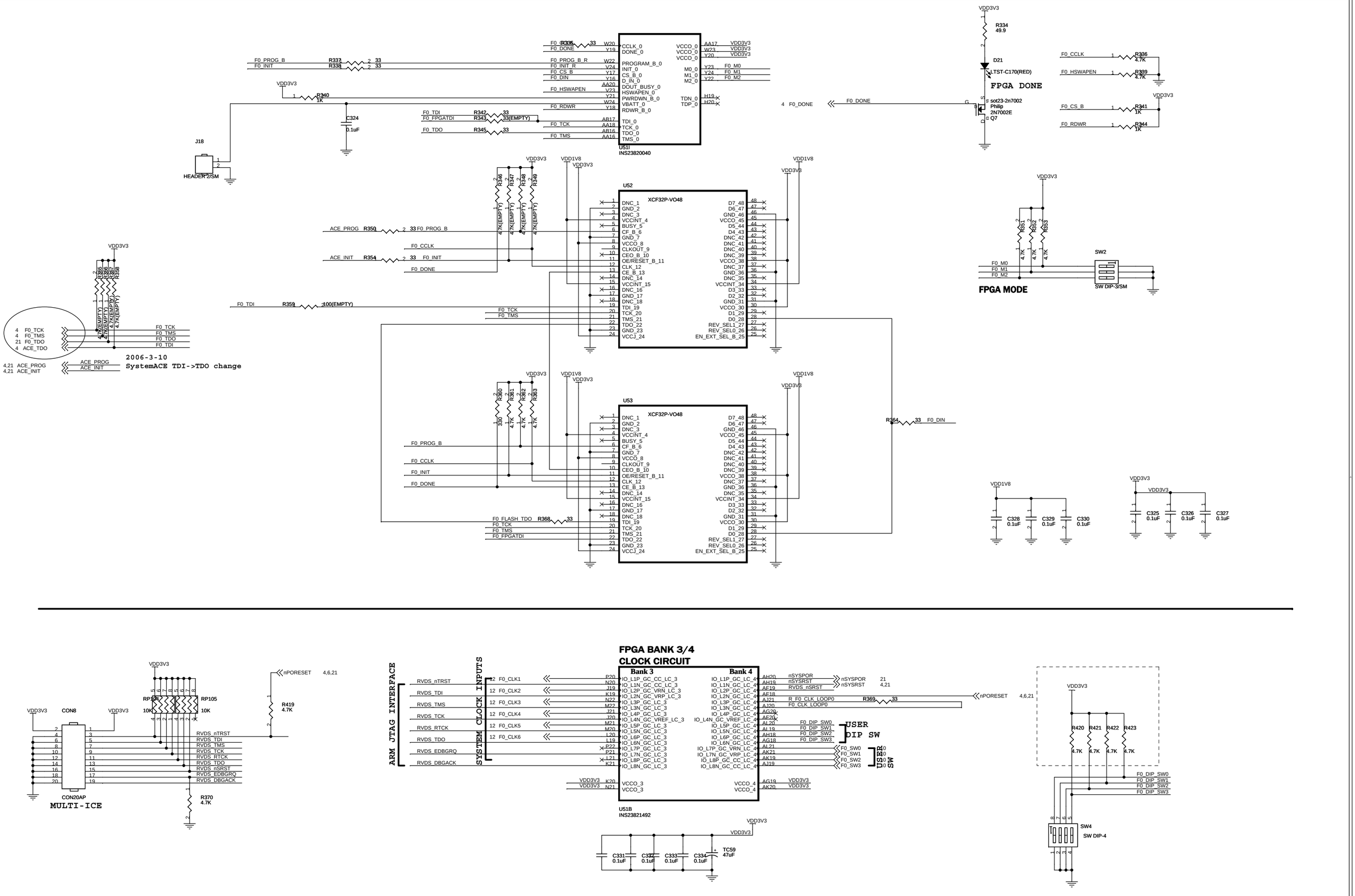
Date: Thursday, August 10, 2006	Sheet 14 of 26	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U52
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U53
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J18
HEADER 2/SM
FPGA DONE
D21
LTST-C170(RED)
Q7
2N7002E
FPGA MODE
SW2
SW DIP-3/SM
2006-3-10
SystemACE TDI->TDO change
FPGA BANK 3/4
CLOCK CIRCUIT
Bank 3
Bank 4
U51B
INS23821492
ARM JTAG INTERFACE
SYSTEM CLOCK INPUTS
USER
DIP SW
USER
SW
CON8
CON20AP
MULTI-ICE
SW4
SW DIP-4
SANGHWA FPGA PLATFORM
FPGA0 CONFIG/CLOCK
Size A2
Document Number
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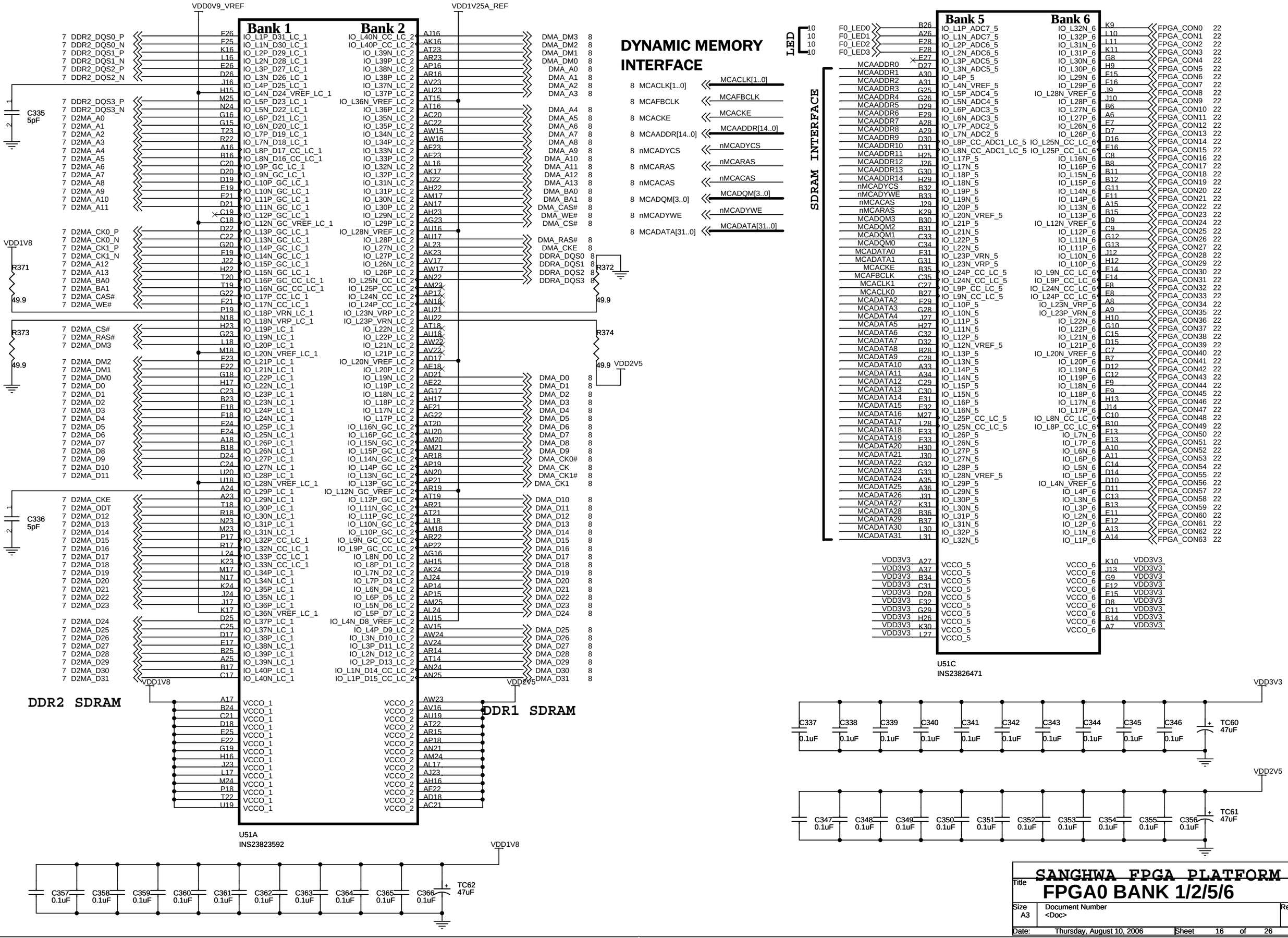
Rev 1.0
Date: Thursday, August 10, 2006
Sheet 15 of 26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Bank 1
Bank 2
Bank 5
Bank 6
DDR2 SDRAM
DDR1 SDRAM
SDRAM INTERFACE
LED
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U51C
INS23826471
VDD0V9_VREF
VDD1V25A_REF
VDD1V8
VDD2V5
VDD3V3
C335 5pF
C336 5pF
R371 49.9
R372 49.9
R373 49.9
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FPGA0 BANK 1/2/5/6
Title
Size A3
Document Number <Doc>
Rev 1.0
Date: Thursday, August 10, 2006
Sheet 16 of 26

FPGA INTERCONNECTION

Bank 7 / Bank 8

23	FPGA_CON384	AT33	IO_L1P_7	IO_L32N_8	AW10	FPGA_CON320	23
23	FPGA_CON385	AR33	IO_L1N_7	IO_L32P_8	AW11	FPGA_CON321	23
23	FPGA_CON386	AJ30	IO_L2P_7	IO_L31N_8	AV14	FPGA_CON322	23
23	FPGA_CON387	AK31	IO_L2N_7	IO_L31P_8	AW14	FPGA_CON323	23
23	FPGA_CON388	AM30	IO_L3P_7	IO_L30N_8	AU10	FPGA_CON324	23
23	FPGA_CON389	AL30	IO_L3N_7	IO_L30P_8	AV10	FPGA_CON325	23
23	FPGA_CON390	AM31	IO_L4P_7	IO_L29N_8	AR13	FPGA_CON326	23
23	FPGA_CON391	AL31	IO_L4N_VREF_7	IO_L29P_8	AT13	FPGA_CON327	23
23	FPGA_CON392	AP24	IO_L5P_7	IO_L28N_VREF_8	AR11	FPGA_CON328	23
23	FPGA_CON393	AR24	IO_L5N_7	IO_L28P_8	AT11	FPGA_CON329	23
23	FPGA_CON394	AP32	IO_L6P_7	IO_L27N_8	AM13	FPGA_CON330	23
23	FPGA_CON395	AN32	IO_L6N_7	IO_L27P_8	AN14	FPGA_CON331	23
23	FPGA_CON396	AN30	IO_L7P_7	IO_L26N_8	AV9	FPGA_CON332	23
23	FPGA_CON397	AP31	IO_L7N_7	IO_L26P_8	AW9	FPGA_CON333	23
23	FPGA_CON398	AK29	IO_L8P_CC_LC_7	IO_L25N_CC_LC_8	AV12	FPGA_CON334	23
23	FPGA_CON399	AJ29	IO_L8N_CC_LC_7	IO_L25P_CC_LC_8	AW12	FPGA_CON335	23
23	FPGA_CON400	AT24	IO_L9P_CC_LC_7	IO_L24N_CC_LC_8	AV7	FPGA_CON336	23
23	FPGA_CON401	AT25	IO_L9N_CC_LC_7	IO_L24P_CC_LC_8	AV8	FPGA_CON337	23
23	FPGA_CON402	AW34	IO_L10P_7	IO_L23N_VRP_8	AU11	FPGA_CON338	23
23	FPGA_CON403	AV34	IO_L10N_7	IO_L23P_VRN_8	AU12	FPGA_CON339	23
23	FPGA_CON404	AW30	IO_L11P_7	IO_L22N_8	AT9	FPGA_CON340	23
23	FPGA_CON405	AW31	IO_L11N_7	IO_L22P_8	AT10	FPGA_CON341	23
23	FPGA_CON406	AV33	IO_L12P_7	IO_L21N_8	AU13	FPGA_CON342	23
23	FPGA_CON407	AU33	IO_L12N_VREF_7	IO_L21P_8	AV13	FPGA_CON343	23
23	FPGA_CON408	AP25	IO_L13P_7	IO_L20N_VREF_8	AP9	FPGA_CON344	23
23	FPGA_CON409	AP26	IO_L13N_7	IO_L20P_8	AR9	FPGA_CON345	23
23	FPGA_CON410	AT31	IO_L14P_7	IO_L19N_8	AP12	FPGA_CON346	23
23	FPGA_CON411	AT30	IO_L14N_7	IO_L19P_8	AR12	FPGA_CON347	23
23	FPGA_CON412	AU25	IO_L15P_7	IO_L18N_8	AT8	FPGA_CON348	23
23	FPGA_CON413	AV25	IO_L15N_7	IO_L18P_8	AU8	FPGA_CON349	23
23	FPGA_CON414	AR31	IO_L16P_7	IO_L17N_8	AL13	FPGA_CON350	23
23	FPGA_CON415	AR32	IO_L16N_7	IO_L17P_8	AL14	FPGA_CON351	23
23	FPGA_CON416	AL26	IO_L17P_7	IO_L16N_8	AU6	FPGA_CON352	23
23	FPGA_CON417	AM27	IO_L17N_7	IO_L16P_8	AU7	FPGA_CON353	23
23	FPGA_CON418	AU31	IO_L18P_7	IO_L15N_8	AP11	FPGA_CON354	23
23	FPGA_CON419	AU32	IO_L18N_7	IO_L15P_8	AN12	FPGA_CON355	23
23	FPGA_CON420	AV28	IO_L19P_7	IO_L14N_8	AJ12	FPGA_CON356	23
23	FPGA_CON421	AU28	IO_L19N_7	IO_L14P_8	AH13	FPGA_CON357	23
23	FPGA_CON422	AW32	IO_L20P_7	IO_L13N_8	AN10	FPGA_CON358	23
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23	FPGA_CON424	AW25	IO_L21P_7	IO_L12N_VREF_8	AL10	FPGA_CON360	23
23	FPGA_CON425	AW26	IO_L21N_7	IO_L12P_8	AM10	FPGA_CON361	23
23	FPGA_CON426	AP29	IO_L22P_7	IO_L11N_8	AW6	FPGA_CON362	23
23	FPGA_CON427	AN29	IO_L22N_7	IO_L11P_8	AW7	FPGA_CON363	23
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23	FPGA_CON432	AP27	IO_L25P_CC_SM7_LC_7	IO_L8N_CC_LC_8	AV3	FPGA_CON368	23
23	FPGA_CON433	AR27	IO_L25N_CC_SM7_LC_7	IO_L8P_CC_LC_8	AV4	FPGA_CON369	23
23	FPGA_CON434	AV30	IO_L26P_SM6_7	IO_L7N_8	AR7	FPGA_CON370	23
23	FPGA_CON435	AU30	IO_L26N_SM6_7	IO_L7P_8	AR8	FPGA_CON371	23
23	FPGA_CON436	AR26	IO_L27P_SM5_7	IO_L6N_8	AN7	FPGA_CON372	23
23	FPGA_CON437	AT26	IO_L27N_SM5_7	IO_L6P_8	AP7	FPGA_CON373	23
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23	FPGA_CON440	AV27	IO_L29P_SM4_7	IO_L4N_VREF_8	AK9	FPGA_CON376	23
23	FPGA_CON441	AW27	IO_L29N_SM4_7	IO_L4P_8	AL9	FPGA_CON377	23
23	FPGA_CON442	AT29	IO_L30P_SM3_7	IO_L3N_8	AN8	FPGA_CON378	23
23	FPGA_CON443	AR29	IO_L30N_SM3_7	IO_L3P_8	AN9	FPGA_CON379	23
23	FPGA_CON444	AU26	IO_L31P_SM2_7	IO_L2N_8	AJ9	FPGA_CON380	23
23	FPGA_CON445	AU27	IO_L31N_SM2_7	IO_L2P_8	AJ10	FPGA_CON381	23
23	FPGA_CON446	AT28	IO_L32P_SM1_7	IO_L1N_8	AU5	FPGA_CON382	23
23	FPGA_CON447	AR28	IO_L32N_SM1_7	IO_L1P_8	AV5	FPGA_CON383	23

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VDD3V3	AN31	VCCO_7	VCCO_8	AV6	VDD3V3
VDD3V3	AP28	VCCO_7	VCCO_8	AU9	VDD3V3
VDD3V3	AR25	VCCO_7	VCCO_8	AT12	VDD3V3
VDD3V3	AT32	VCCO_7	VCCO_8	AP8	VDD3V3
VDD3V3	AU29	VCCO_7	VCCO_8	AN11	VDD3V3
VDD3V3	AV26	VCCO_7	VCCO_8	AM14	VDD3V3
VDD3V3	AW33	VCCO_7	VCCO_8	AK10	VDD3V3
			VCCO_8	AJ13	VDD3V3

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Bank 9 / Bank 10

14	EXP_IO_64	D34	IO_L1P_9	IO_L32N_10	T11	FPGA_CON64	23
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14	EXP_IO_68	E34	IO_L3P_9	IO_L30N_10	P7	FPGA_CON68	23
14	EXP_IO_69	E34	IO_L3N_9	IO_L30P_10	N7	FPGA_CON69	23
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14	EXP_IO_71	G35	IO_L4N_VREF_9	IO_L29P_10	L6	FPGA_CON71	23
14	EXP_IO_72	D36	IO_L5P_9	IO_L28N_VREF_10	N8	FPGA_CON72	23
14	EXP_IO_73	D37	IO_L5N_9	IO_L28P_10	P9	FPGA_CON73	23
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14	EXP_IO_75	H34	IO_L6N_9	IO_L27P_10	M8	FPGA_CON75	23
14	EXP_IO_76	E36	IO_L7P_9	IO_L26N_10	U13	FPGA_CON76	23
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14	EXP_IO_99	H38	IO_L18N_9	IO_L15P_10	E4	FPGA_CON99	23
14	EXP_IO_100	N30	IO_L19P_9	IO_L14N_10	N10	FPGA_CON100	23
14	EXP_IO_101	P29	IO_L19N_9	IO_L14P_10	M10	FPGA_CON101	23
14	EXP_IO_102	L34	IO_L20P_9	IO_L13N_10	T15	FPGA_CON102	23
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14	EXP_IO_104	J36	IO_L21P_9	IO_L12N_VREF_10	F3	FPGA_CON104	23
14	EXP_IO_105	J37	IO_L21N_9	IO_L12P_10	F4	FPGA_CON105	23
14	EXP_IO_106	K36	IO_L22P_9	IO_L11N_10	N12	FPGA_CON106	23
14	EXP_IO_107	L36	IO_L22N_9	IO_L11P_10	M11	FPGA_CON107	23
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14	EXP_IO_109	J39	IO_L23N_VRP_9	IO_L10P_10	F5	FPGA_CON109	23
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14	EXP_IO_111	N32	IO_L24N_CC_LC_9	IO_L9P_CC_LC_10	G7	FPGA_CON111	23
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14	EXP_IO_114	M35	IO_L26P_9	IO_L7N_10	F6	FPGA_CON114	23
14	EXP_IO_115	N35	IO_L26N_9	IO_L7P_10	E6	FPGA_CON115	23
14	EXP_IO_116	K37	IO_L27P_9	IO_L6N_10	C2	FPGA_CON116	23
14	EXP_IO_117	K38	IO_L27N_9	IO_L6P_10	C3	FPGA_CON117	23
14	EXP_IO_118	N33	IO_L28P_9	IO_L5N_10	D5	FPGA_CON118	23
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14	EXP_IO_120	P31	IO_L29P_9	IO_L4N_VREF_10	D4	FPGA_CON120	23
14	EXP_IO_121	P32	IO_L29N_9	IO_L4P_10	C4	FPGA_CON121	23
14	EXP_IO_122	K39	IO_L30P_9	IO_L3N_10	C5	FPGA_CON122	23
14	EXP_IO_123	L39	IO_L30N_9	IO_L3P_10	B5	FPGA_CON123	23
14	EXP_IO_124	L38	IO_L31P_9	IO_L2N_10	B3	FPGA_CON124	23
14	EXP_IO_125	M38	IO_L31N_9	IO_L2P_10	A3	FPGA_CON125	23
14	EXP_IO_126	M36	IO_L32P_9	IO_L1N_10	A4	FPGA_CON126	23
14	EXP_IO_127	M37	IO_L32N_9	IO_L1P_10	A5	FPGA_CON127	23

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VDD3V3	E35	VCCO_9	VCCO_10	R15	VDD3V3
VDD3V3	G39	VCCO_9	VCCO_10	P8	VDD3V3
VDD3V3	H36	VCCO_9	VCCO_10	N11	VDD3V3
VDD3V3	J33	VCCO_9	VCCO_10	L7	VDD3V3
VDD3V3	L37	VCCO_9	VCCO_10	H6	VDD3V3
VDD3V3	M34	VCCO_9	VCCO_10	F2	VDD3V3
VDD3V3	N31	VCCO_9	VCCO_10	E5	VDD3V3
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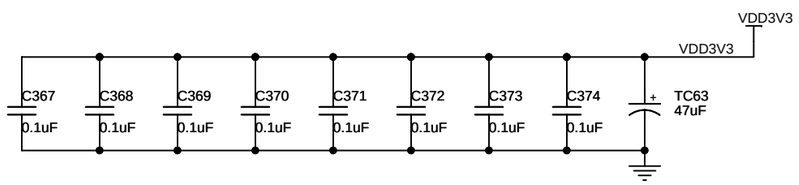
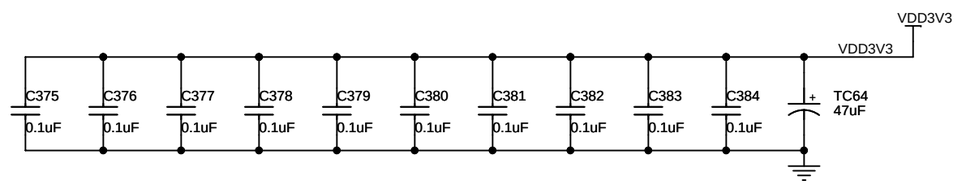




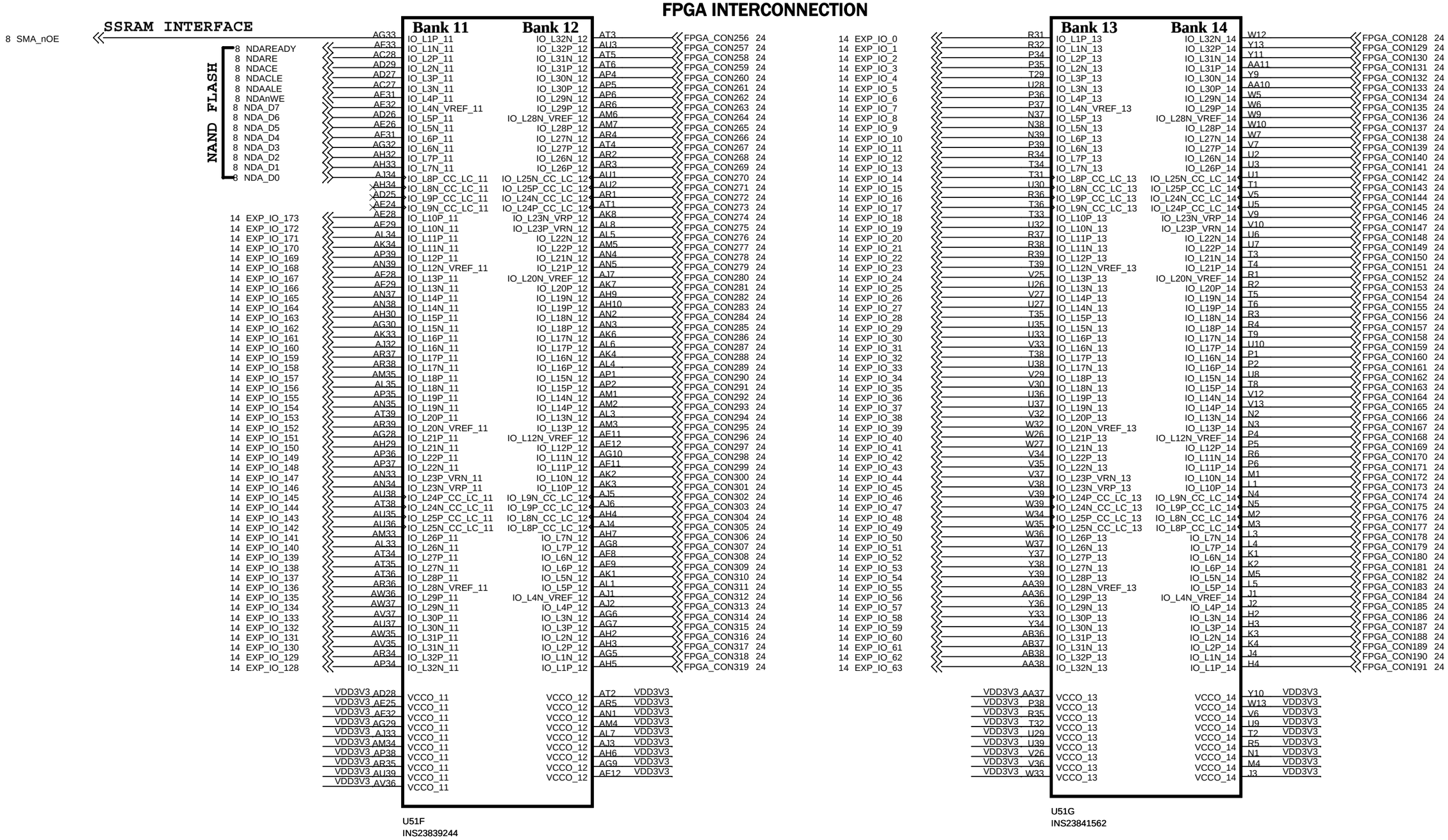
SANGHWA FPGA PLATFORM

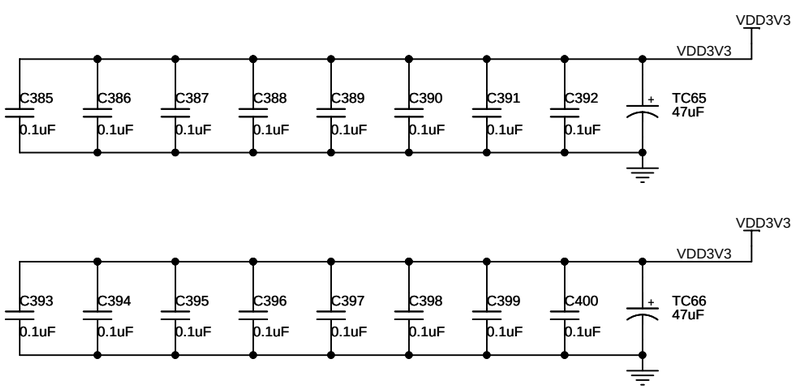
Title: FPGA0 BANK 7/8/9/10

Size	Document Number	Rev
A3	<Doc>	1.0

Date: Thursday, August 10, 2006 Sheet 17 of 26

FPGA INTERCONNECTION





Title	SANGHWA FPGA PLATFORM FPGA0 BANK 11/12/13/14		
Size A3	Document Number <Doc>		Rev 1.0
Date: Thursday, August 10, 2006	Sheet 18 of 26		

VIRTEX-4 FPGA
POWER GROUND

GND	D13	GND	J28
GND	K13	GND	N28
GND	M13	GND	W28
GND	P13	GND	AJ28
GND	AD13	GND	AW28
GND	AF13	GND	B29
GND	AK13	GND	M29
GND	AP13	GND	AB29
GND	G14	GND	AM29
GND	L14	GND	E30
GND	N14	GND	R30
GND	U14	GND	AF30
GND	W14	GND	AR30
GND	AF14	GND	H31
GND	AG14	GND	V31
GND	AJ14	GND	AH31
GND	AU14	GND	AV31
GND	K15	GND	A32
GND	M15	GND	L32
GND	P15	GND	AA32
GND	V15	GND	AL32
GND	Y15	GND	D33
GND	AD15	GND	P33
GND	AF15	GND	AD33
GND	AK15	GND	AP33
GND	C16	GND	G34
GND	N16	GND	U34
GND	U16	GND	AG34
GND	W16	GND	AU34
GND	AC16	GND	K35
GND	AE16	GND	Y35
GND	AN16	GND	AK35
GND	F17	GND	C36
GND	T17	GND	N36
GND	V17	GND	AC36
GND	AF17	GND	AN36
GND	AT17	GND	F37
GND	J18	GND	T37
GND	W18	GND	AF37
GND	AC18	GND	AT37
GND	AJ18	GND	A38
GND	AW18	GND	J38
GND	B19	GND	W38
GND	M19	GND	AJ38
GND	V19	GND	AW38
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GND	AD19	GND	M39
GND	AM19	GND	AB39
GND	E20	GND	AM39
GND	R20	GND	AV39
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GND	AR20	GND	H1
GND	H21	GND	V1
GND	T21	GND	AH1
GND	V21	GND	AV1
GND	AB21	GND	A2
GND	AH21	GND	L2
GND	AV21	GND	AA2
GND	A22	GND	AL2
GND	L22	GND	AW2
GND	U22	GND	D3
GND	AA22	GND	P3
GND	AL22	GND	AD3
GND	D23	GND	AP3
GND	P23	GND	G4
GND	AB23	GND	U4
GND	AD23	GND	AG4
GND	AP23	GND	AU4
GND	G24	GND	K5
GND	R24	GND	Y5
GND	U24	GND	AK5
GND	AA24	GND	C6
GND	AC24	GND	N6
GND	AG24	GND	AC6
GND	AU24	GND	AN6
GND	K25	GND	F7
GND	P25	GND	T7
GND	T25	GND	AF7
GND	Y25	GND	AT7
GND	AB25	GND	J8
GND	AF25	GND	W8
GND	AH25	GND	AJ8
GND	AK25	GND	AW8
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GND	L26	GND	M9
GND	N26	GND	AB9
GND	R26	GND	AM9
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GND	AJ26	GND	AR10
GND	AN26	GND	H11
GND	F27	GND	V11
GND	K27	GND	AH11
GND	P27	GND	AV11
GND	T27	GND	A12
GND	AF27	GND	L12
GND	AH27	GND	AA12
GND	AK27	GND	AG12
GND	AT27	GND	AL12

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SSRAM INTERFACE

8	SMA_A2	Y27	IO_L1P_15	IO_L32N_16	AF3	FPGA_CON192	25
8	SMA_A3	AA28	IO_L1N_15	IO_L32P_16	AG3	FPGA_CON193	25
8	SMA_A4	W29	IO_L2P_15	IO_L31N_16	AF8	FPGA_CON194	25
8	SMA_A5	W30	IO_L2N_15	IO_L31P_16	AF9	FPGA_CON195	25
8	SMA_A6	AA34	IO_L3P_15	IO_L30N_16	AG1	FPGA_CON196	25
8	SMA_A7	AA35	IO_L3N_15	IO_L30P_16	AG2	FPGA_CON197	25
8	SMA_A8	Y29	IO_L4P_15	IO_L29N_16	AF5	FPGA_CON198	25
8	SMA_A9	AA30	IO_L4N_VREF_15	IO_L29P_16	AF6	FPGA_CON199	25
8	SMA_A10	AC38	IO_L5P_15	IO_L28N_VREF_16	AD6	FPGA_CON200	25
8	SMA_A11	AC39	IO_L5N_15	IO_L28P_16	AE6	FPGA_CON201	25
8	SMA_A12	AA31	IO_L6P_15	IO_L27N_16	AE4	FPGA_CON202	25
8	SMA_A13	Y31	IO_L6N_15	IO_L27P_16	AE4	FPGA_CON203	25
8	SMA_A14	AC35	IO_L7P_15	IO_L26N_16	AE1	FPGA_CON204	25
8	SMA_A15	AB35	IO_L7N_15	IO_L26P_16	AE1	FPGA_CON205	25
8	SMA_A16	AB33	IO_L8P_CC_LC_15	IO_L25N_CC_LC_16	AD11	FPGA_CON206	25
8	SMA_A17	AA33	IO_L8N_CC_LC_15	IO_L25P_CC_LC_16	AC12	FPGA_CON207	25
8	SMA_A18	AC33	IO_L9P_CC_LC_15	IO_L24N_CC_LC_16	AE2	FPGA_CON208	25
8	SMA_A19	AC34	IO_L9N_CC_LC_15	IO_L24P_CC_LC_16	AE3	FPGA_CON209	25
8	SMA_A20	AD37	IO_L10P_15	IO_L23N_VRP_16	AC7	FPGA_CON210	25
8	SMA_nADSC	AC37	IO_L10N_15	IO_L23P_VRN_16	AD7	FPGA_CON211	25
8	SMA_nADSP	AC32	IO_L11P_15	IO_L22N_16	AD9	FPGA_CON212	25
8	SMA_nADV	AB31	IO_L11N_15	IO_L22P_16	AC10	FPGA_CON213	25
8	SMA_nBW1	AE39	IO_L12P_15	IO_L21N_16	AC14	FPGA_CON214	25
8	SMA_nBW2	AD39	IO_L12N_VREF_15	IO_L21P_16	AB15	FPGA_CON215	25
8	SMA_nBW3	AE38	IO_L13P_15	IO_L20N_VREF_16	AC13	FPGA_CON216	25
8	SMA_nBW4	AE39	IO_L13N_15	IO_L20P_16	AB13	FPGA_CON217	25
8	SMA_nBWE	AD35	IO_L14P_15	IO_L19N_16	AD4	FPGA_CON218	25
8	SMA_nCE	AD36	IO_L14N_15	IO_L19P_16	AD5	FPGA_CON219	25
8	SMA_CLK	AC30	IO_L15P_15	IO_L18N_16	AD1	FPGA_CON220	25
8	SMA_nCS1	AB30	IO_L15N_15	IO_L18P_16	AD2	FPGA_CON221	25
8	SMA_nGW	AE37	IO_L16P_15	IO_L17N_16	AB8	FPGA_CON222	25
8	SMA_nLBO	AE38	IO_L16N_15	IO_L17P_16	AC8	FPGA_CON223	25
8	SMA_D31	AB27	IO_L17P_15	IO_L16N_16	AC4	FPGA_CON224	25
8	SMA_D30	AB28	IO_L17N_15	IO_L16P_16	AC5	FPGA_CON225	25
8	SMA_D29	AE34	IO_L18P_15	IO_L15N_16	AB10	FPGA_CON226	25
8	SMA_D28	AD34	IO_L18N_15	IO_L15P_16	AB11	FPGA_CON227	25
8	SMA_D27	AD31	IO_L19P_15	IO_L14N_16	AA8	FPGA_CON228	25
8	SMA_D26	AD32	IO_L19N_15	IO_L14P_16	AB7	FPGA_CON229	25
8	SMA_D25	AE36	IO_L20P_15	IO_L13N_16	AC2	FPGA_CON230	25
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8	SMA_D23	AJ39	IO_L21P_15	IO_L12N_VREF_16	AB1	FPGA_CON232	25
8	SMA_D22	AH39	IO_L21N_15	IO_L12P_16	AB2	FPGA_CON233	25
8	SMA_D21	AG37	IO_L22P_15	IO_L11N_16	AB5	FPGA_CON234	25
8	SMA_D20	AG38	IO_L22N_15	IO_L11P_16	AB6	FPGA_CON235	25
8	SMA_D19	AH37	IO_L23P_VRN_15	IO_L10N_16	AA5	FPGA_CON236	25
8	SMA_D18	AH38	IO_L23N_VRP_15	IO_L10P_16	AA6	FPGA_CON237	25
8	SMA_D17	AE34	IO_L24P_CC_LC_15	IO_L9N_CC_LC_16	AA13	FPGA_CON238	25
8	SMA_D16	AE35	IO_L24N_CC_LC_15	IO_L9P_CC_LC_16	AA14	FPGA_CON239	25
8	SMA_D15	AJ36	IO_L25P_CC_LC_15	IO_L8N_CC_LC_16	AA1	FPGA_CON240	25
8	SMA_D14	AJ37	IO_L25N_CC_LC_15	IO_L8P_CC_LC_16	Y1	FPGA_CON241	25
8	SMA_D13	AG35	IO_L26P_15	IO_L7N_16	AB3	FPGA_CON242	25
8	SMA_D12	AG36	IO_L26N_15	IO_L7P_16	AA3	FPGA_CON243	25
8	SMA_D11	AJ35	IO_L27P_15	IO_L6N_16	Y2	FPGA_CON244	25
8	SMA_D10	AH35	IO_L27N_15	IO_L6P_16	Y3	FPGA_CON245	25
8	SMA_D9	AK38	IO_L28P_15	IO_L5N_16	AA4	FPGA_CON246	25
8	SMA_D8	AK39	IO_L28N_VREF_15	IO_L5P_16	Y4	FPGA_CON247	25
8	SMA_D7	AM37	IO_L29P_15	IO_L4N_VREF_16	Y6	FPGA_CON248	25
8	SMA_D6	AM38	IO_L29N_15	IO_L4P_16	Y7	FPGA_CON249	25
8	SMA_D5	AL38	IO_L30P_15	IO_L3N_16	W1	FPGA_CON250	25
8	SMA_D4	AL39	IO_L30N_15	IO_L3P_16	W2	FPGA_CON251	25
8	SMA_D3	AM36	IO_L31P_15	IO_L2N_16	V2	FPGA_CON252	25
8	SMA_D2	AL36	IO_L31N_15	IO_L2P_16	V3	FPGA_CON253	25
8	SMA_D1	AK36	IO_L32P_15	IO_L1N_16	W4	FPGA_CON254	25
8	SMA_D0	AK37	IO_L32N_15	IO_L1P_16	V4	FPGA_CON255	25

Bank 15 | Bank 16

VDD3V3	AA27	VCCO_15	VCCO_16	W3	VDD3V3
VDD3V3	AB34	VCCO_15	VCCO_16	AF2	VDD3V3
VDD3V3	AC31	VCCO_15	VCCO_16	AF5	VDD3V3
VDD3V3	AD38	VCCO_15	VCCO_16	AD8	VDD3V3
VDD3V3	AE35	VCCO_15	VCCO_16	AC11	VDD3V3
VDD3V3	AG39	VCCO_15	VCCO_16	AC1	VDD3V3
VDD3V3	AH36	VCCO_15	VCCO_16	AB4	VDD3V3
VDD3V3	AL37	VCCO_15	VCCO_16	AB14	VDD3V3
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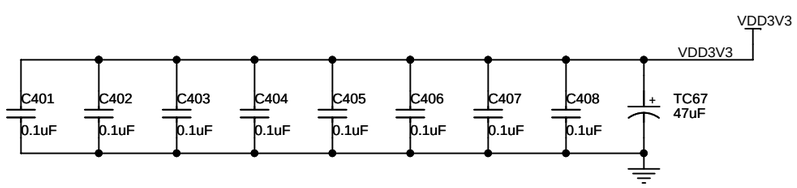

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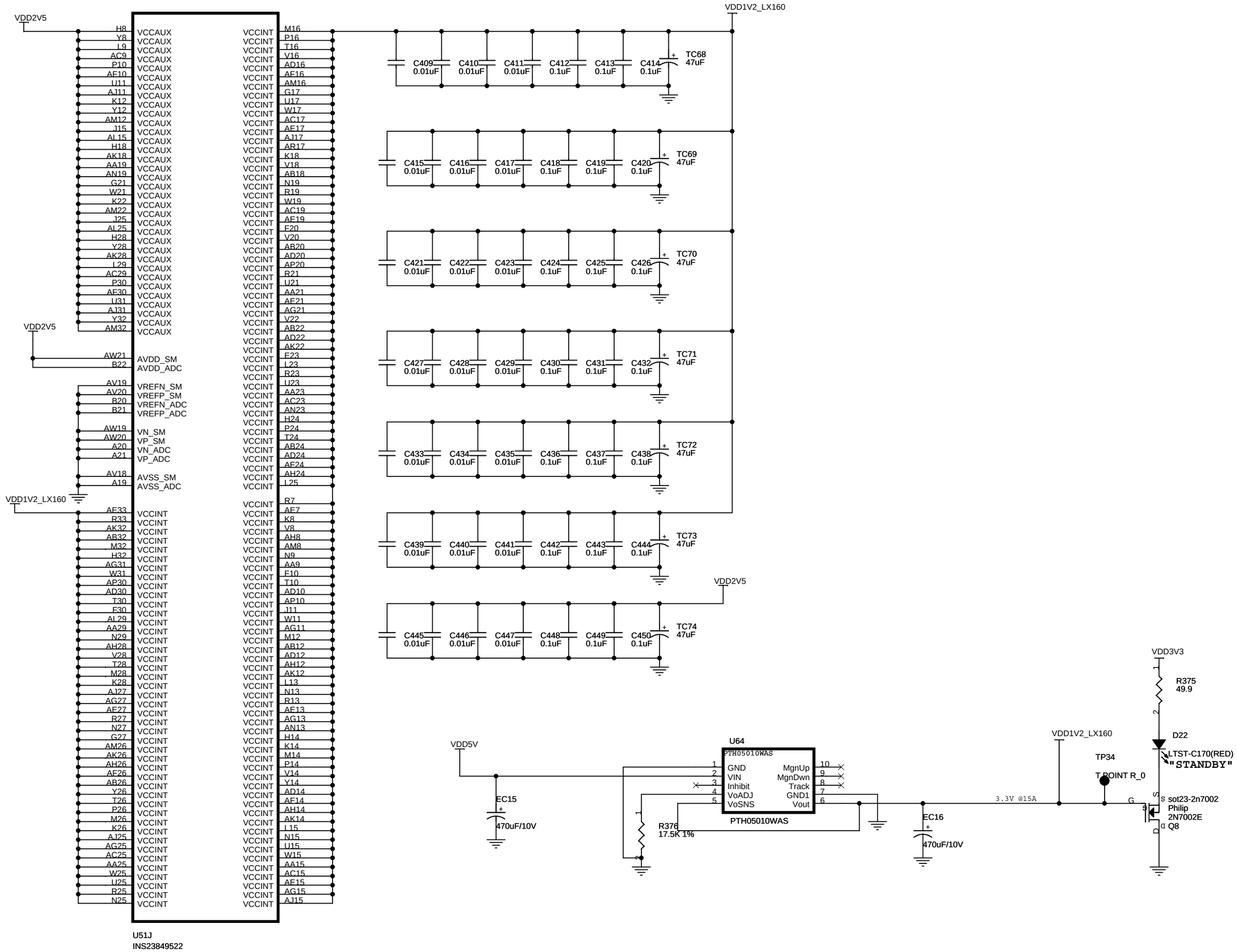
Date: Thursday, August 10, 2006	Sheet 19 of 26		



SANGHWA FPGA PLATFORM

Title
FPGA0 POWER VCCO

Size	Document Number	Rev
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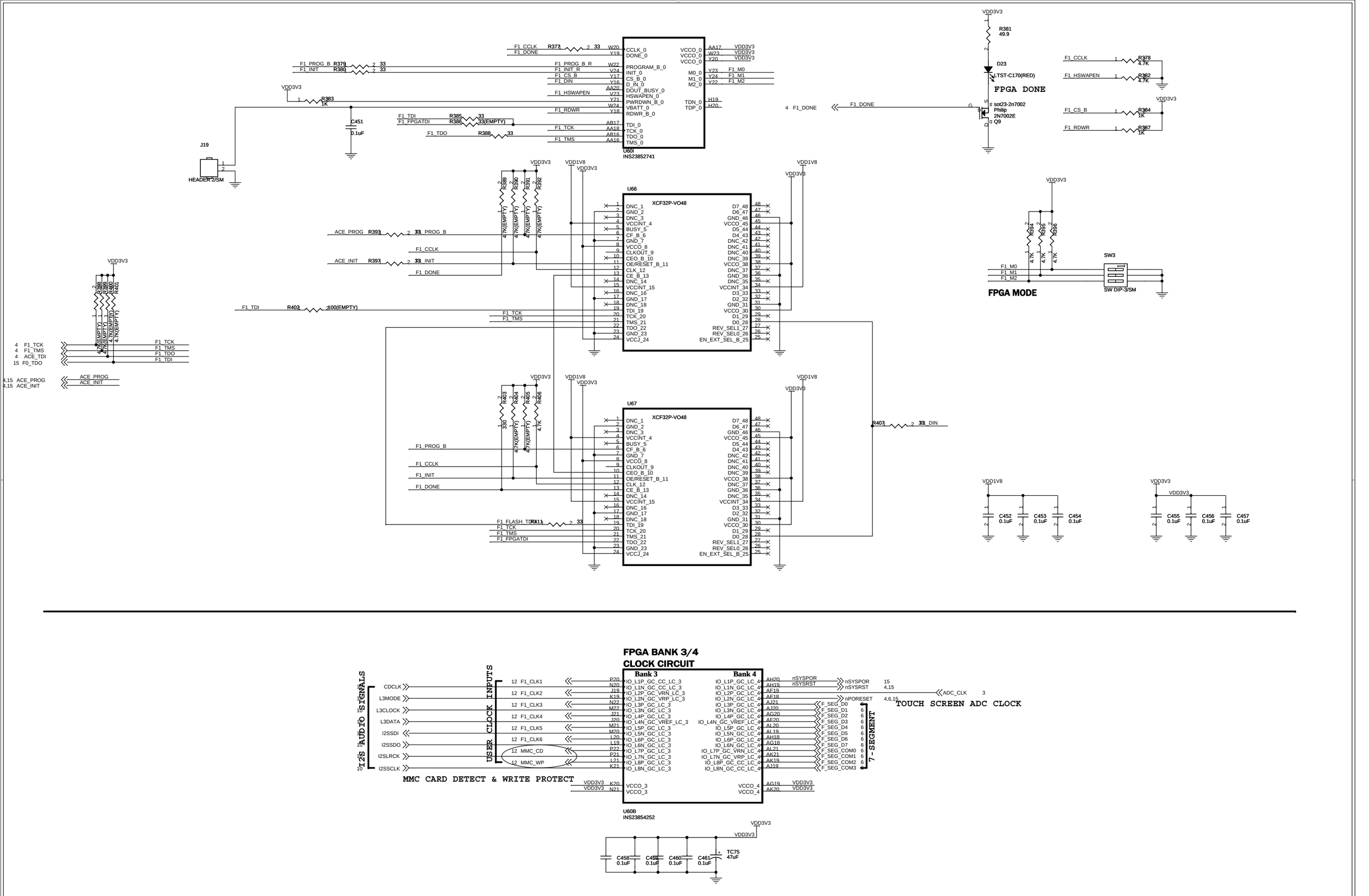
Date: Thursday, August 10, 2006 Sheet 20 of 26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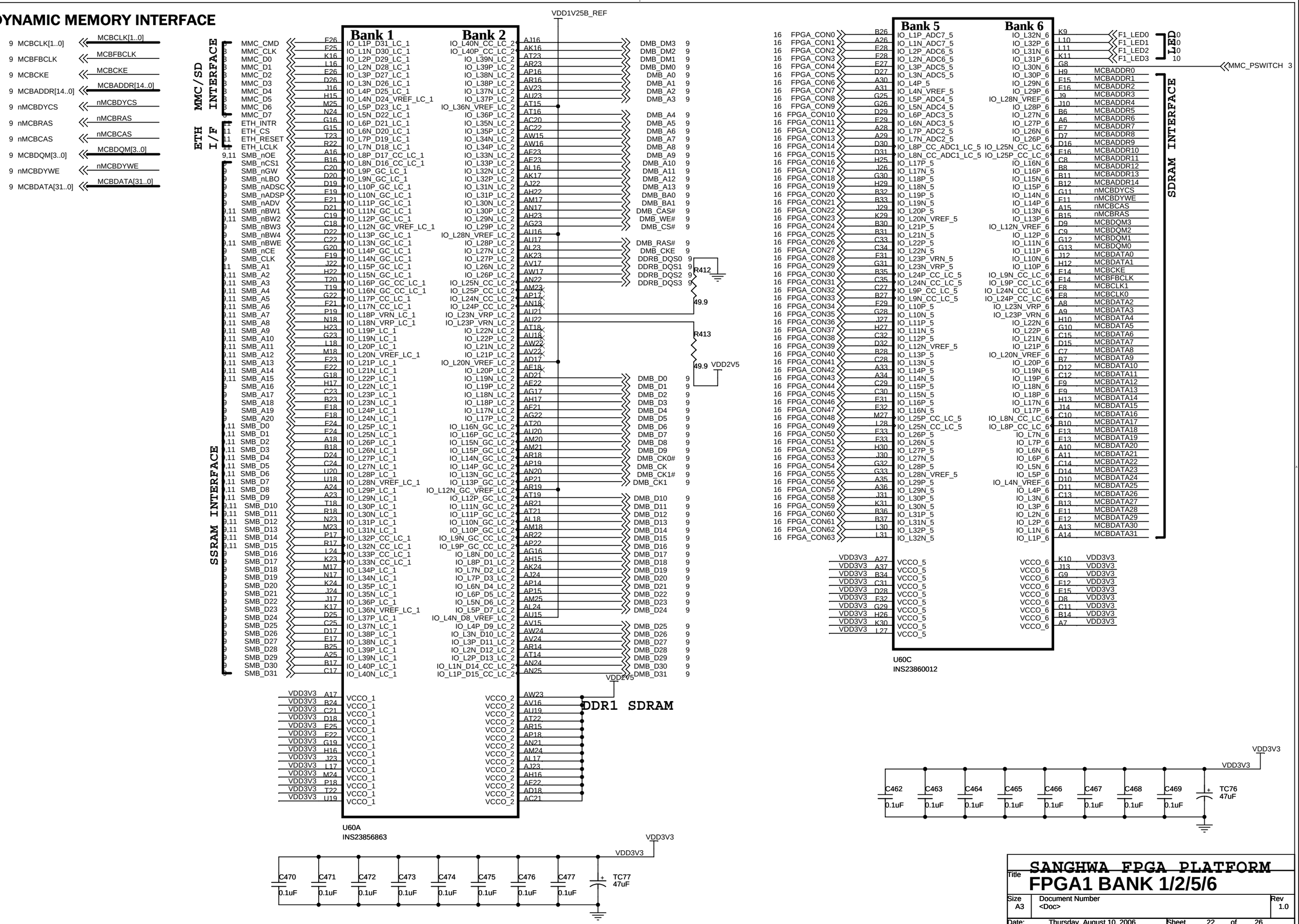
U66
XCF32P-VO48
U67
XCF32P-VO48
FPGA DONE
FPGA MODE
SW3
SW DIP-3/SM
J19
HEADER 2/SM
FPGA BANK 3/4
CLOCK CIRCUIT
Bank 3
Bank 4
U60B
INS23854252
USER CLOCK INPUTS
I2S AUDIO SIGNALS
7-SEGMENT
MMC CARD DETECT & WRITE PROTECT
TOUCH SCREEN ADC CLOCK
SANGHWA FPGA PLATFORM
FPGA1 CONFIG/CLOCK
Size A2
Document Number <Doc>
Rev 1.0
Date: Thursday, August 10, 2006
Sheet 21 of 26

DYNAMIC MEMORY INTERFACE



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FPGA1 BANK 1/2/5/6

Size	Document Number	Rev
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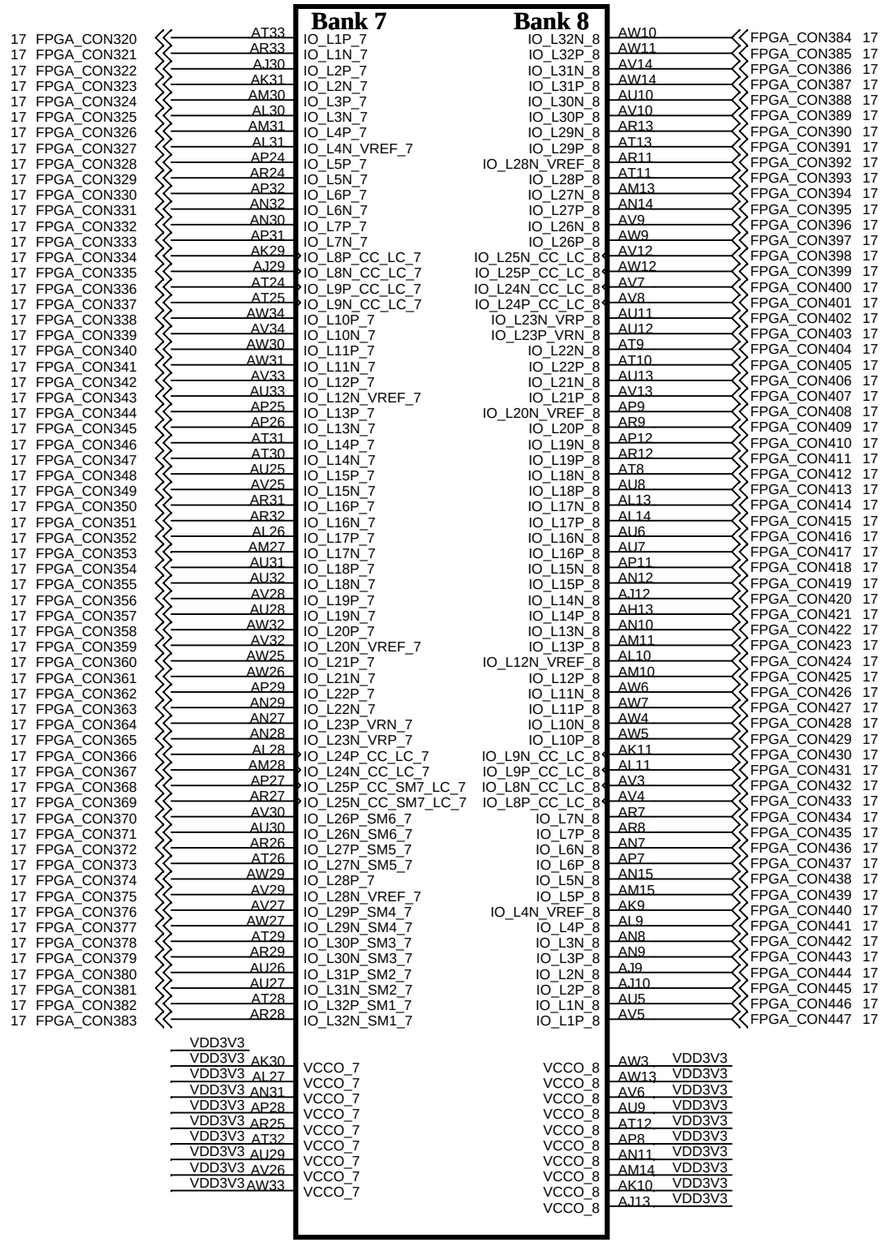
Date: Thursday, August 10, 2006 Sheet 22 of 26



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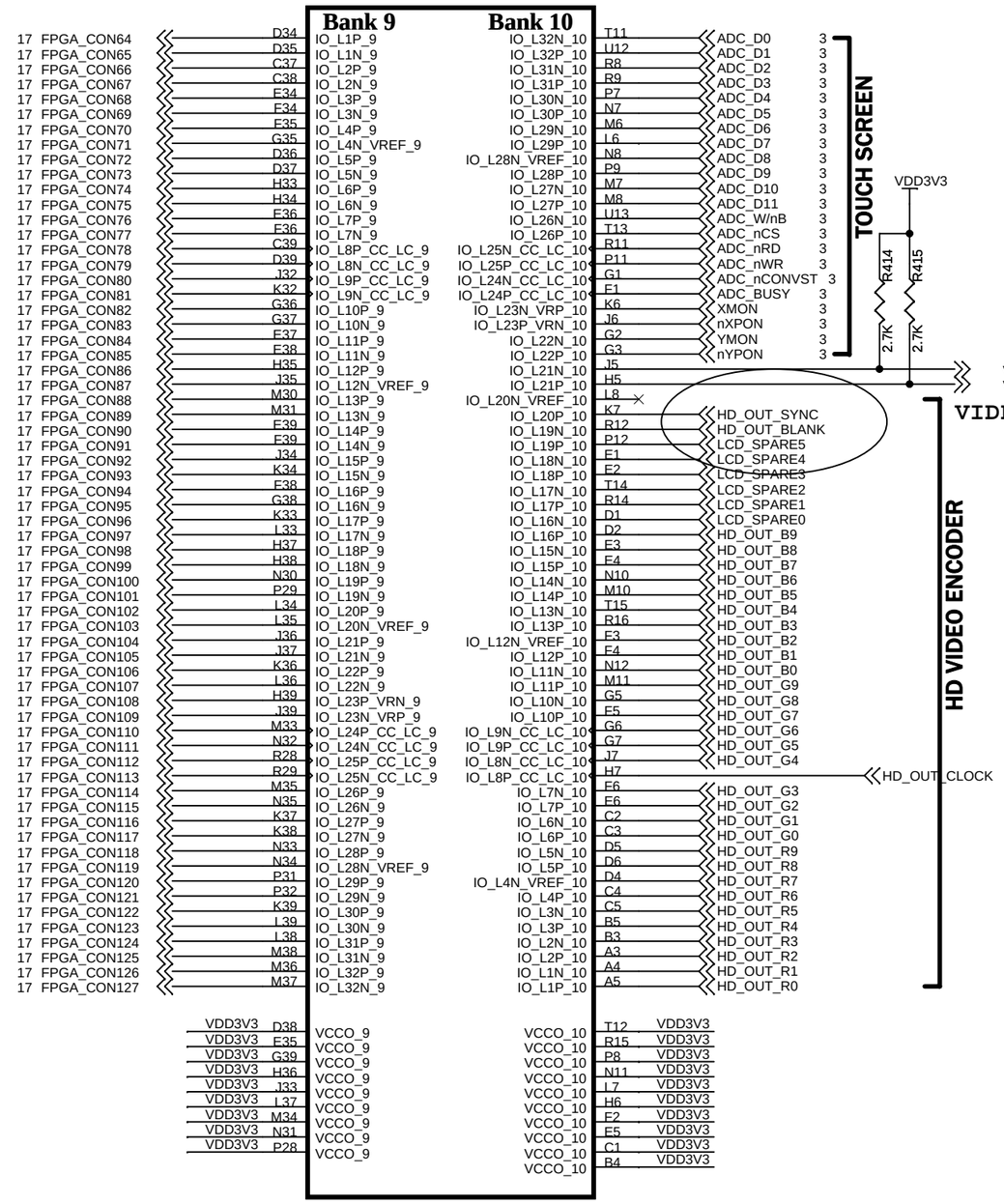

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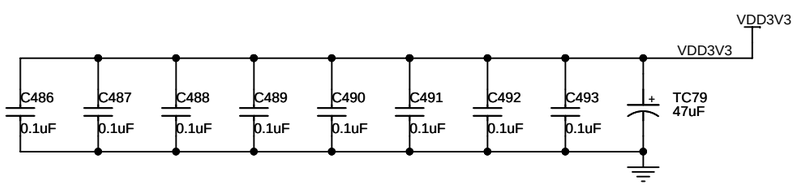

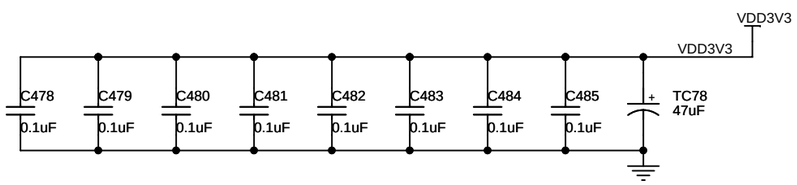

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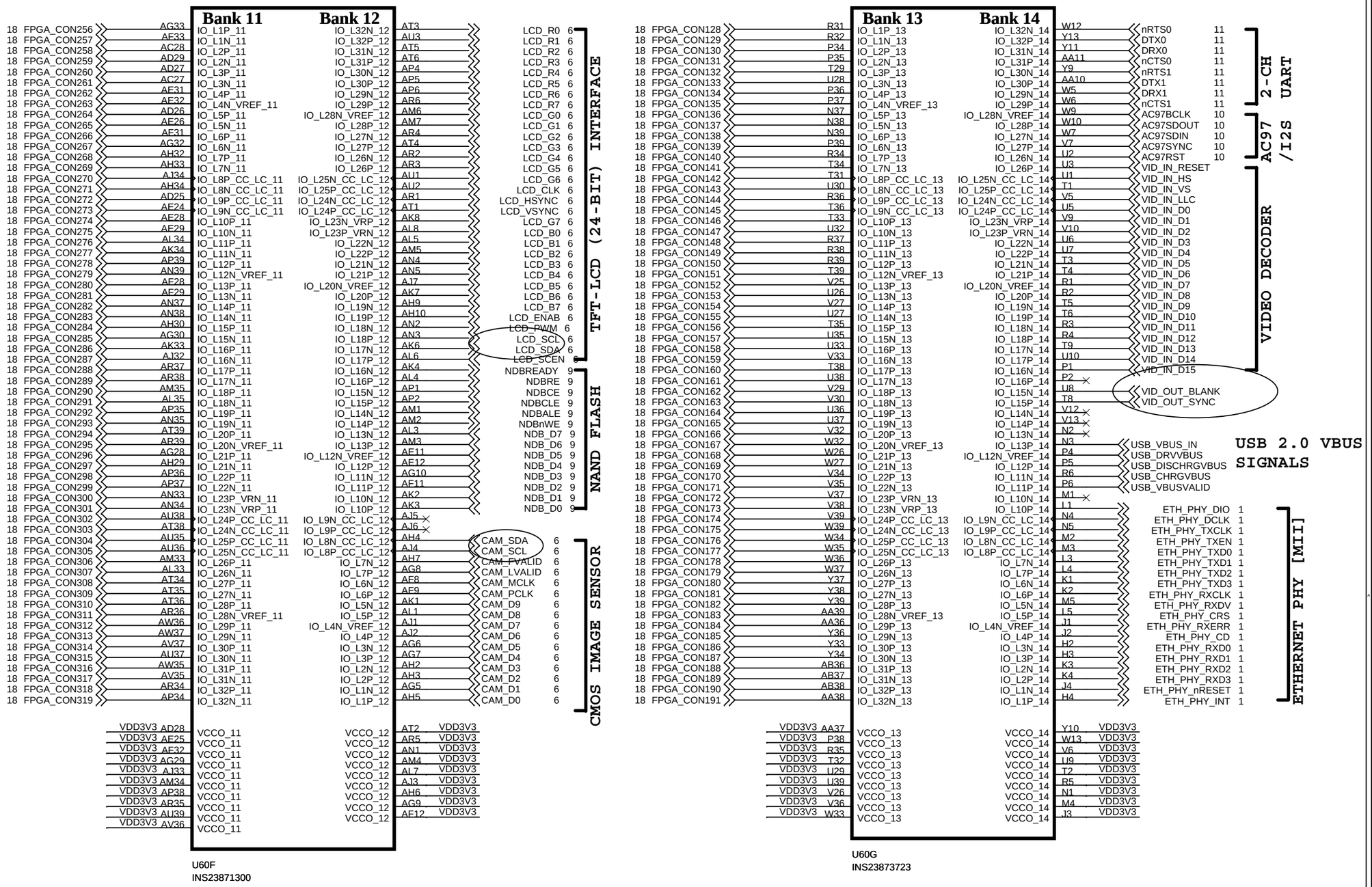
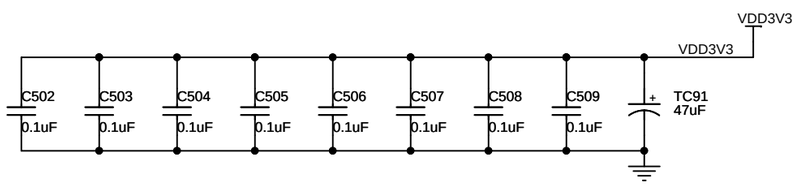
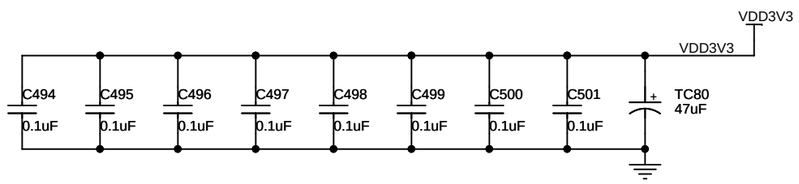
Size A3	Document Number <Doc>	Rev 1.0
Date: Thursday, August 10, 2006	Sheet 23 of 26	







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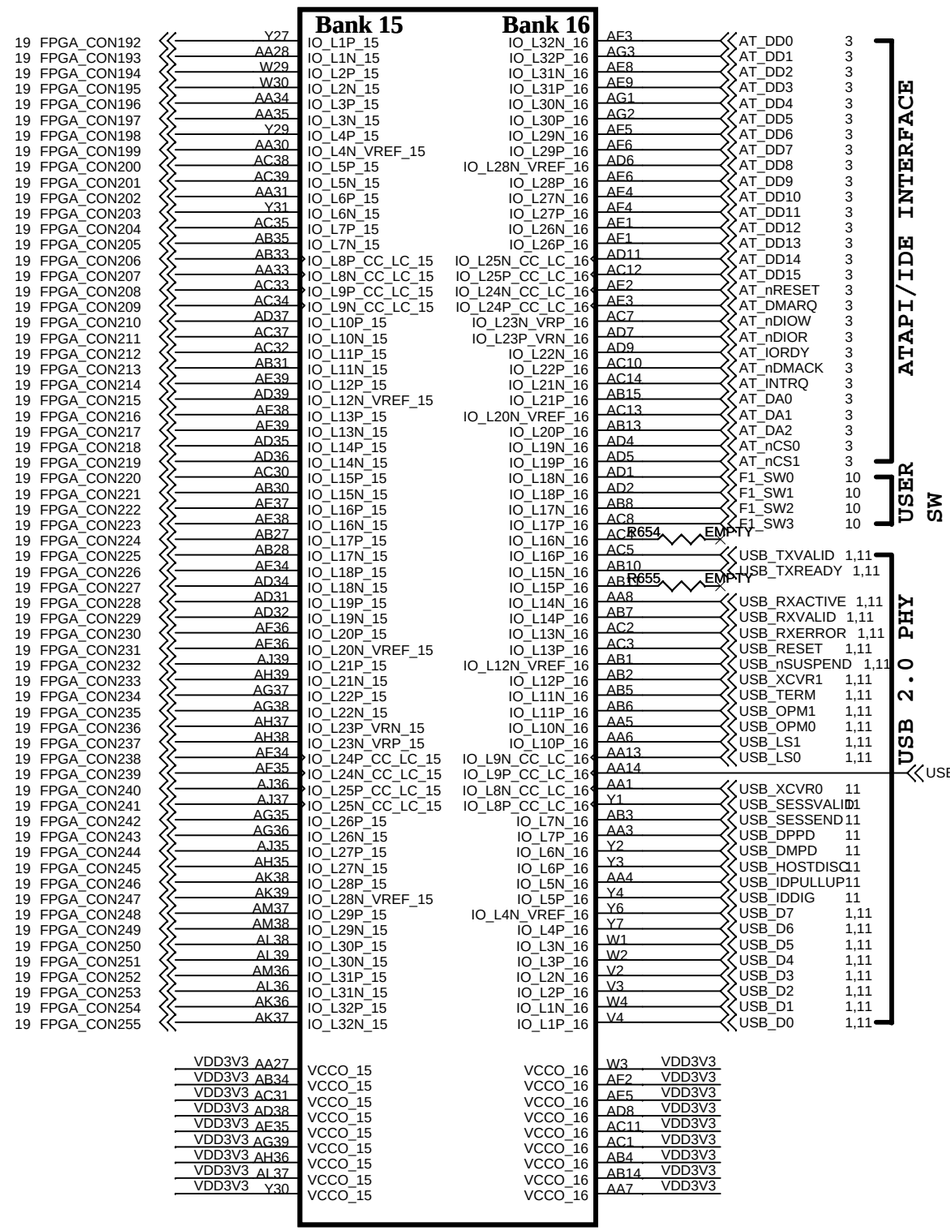
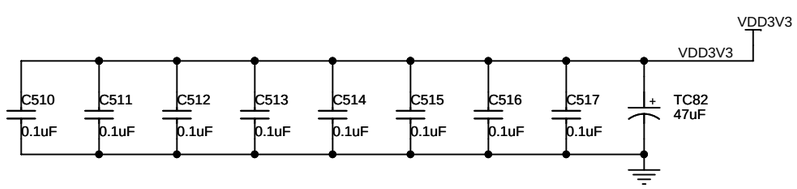
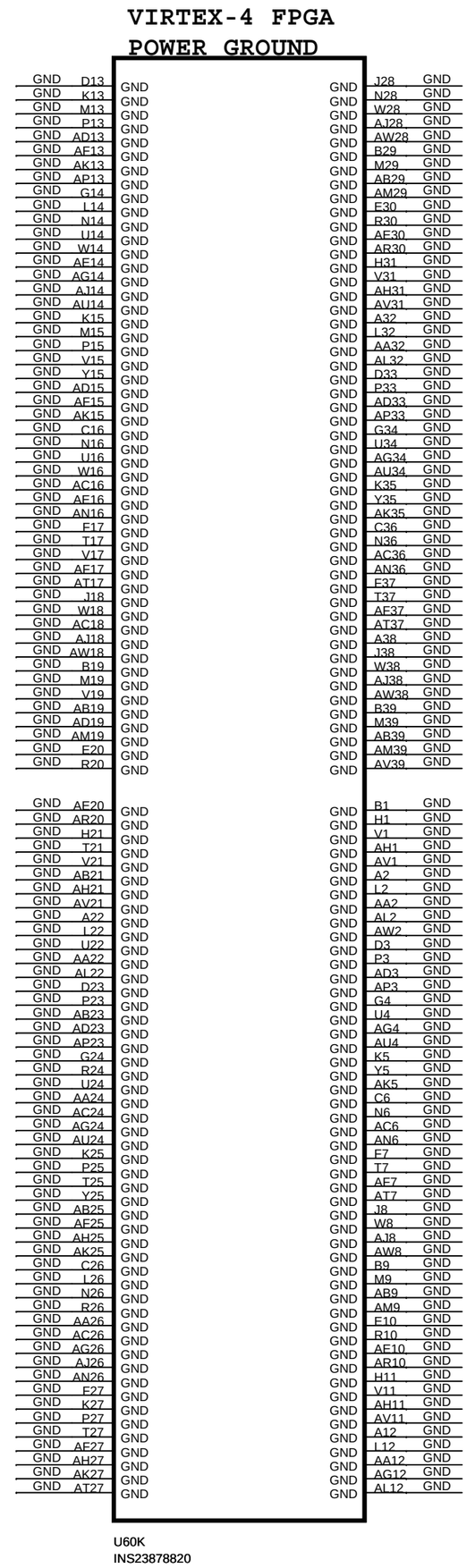
Size A3	Document Number <Doc>
Rev	1.0
Date: Thursday, August 10, 2006	Sheet 24 of 26







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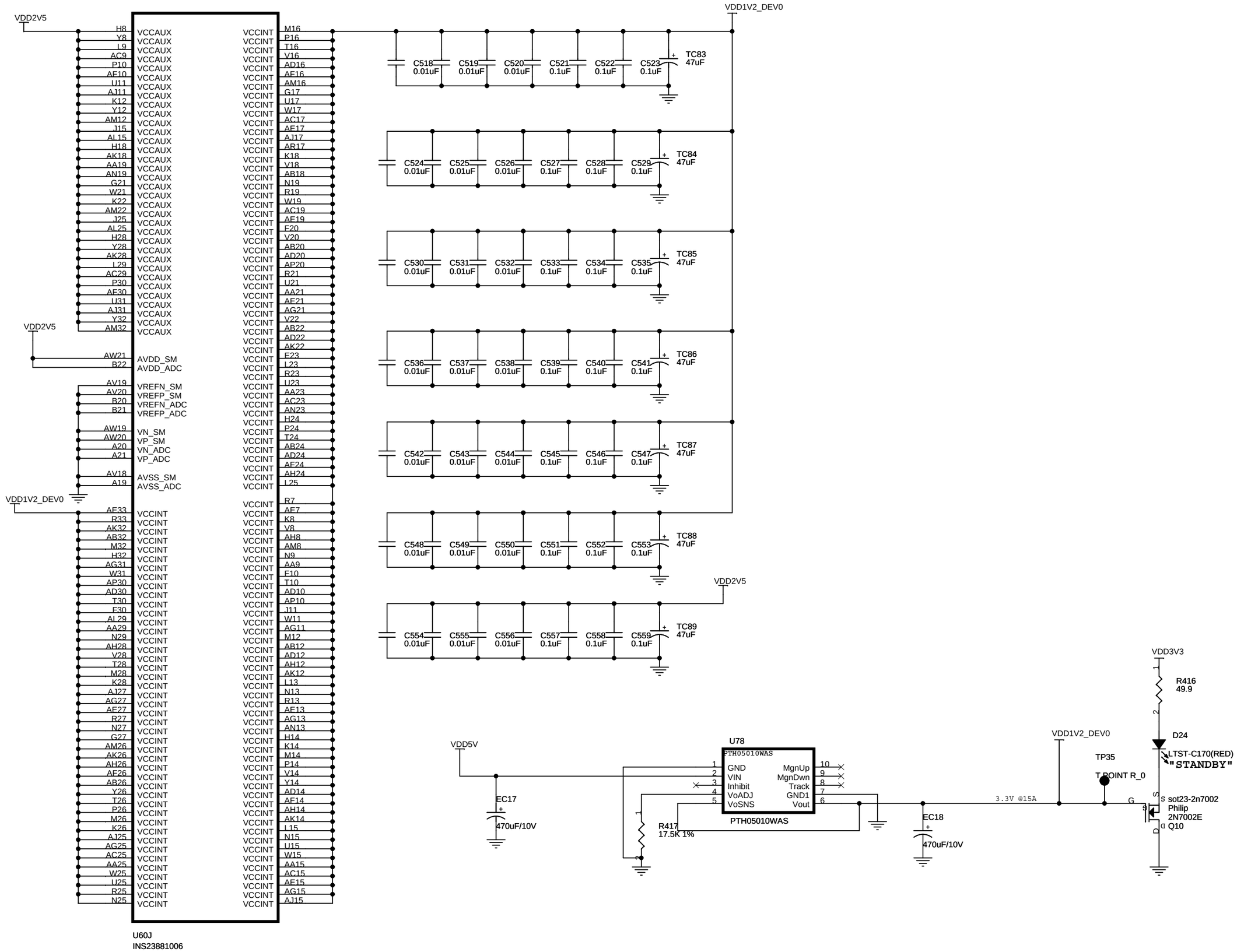
Size A3	Document Number <Doc>	Rev 1.0
Date: Thursday, August 10, 2006	Sheet 25 of 26	



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Size A3	Document Number <Doc>	Rev 1.0
Date: Thursday, August 10, 2006	Sheet 26 of 26	